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PHYSICS

1) A particle moves 3 m north, then 4 m east, and then 12 m vertically upwards, its displacement is :-

(1) 12 m
(2) 13 m
(3) 19 m
(4) 15 m

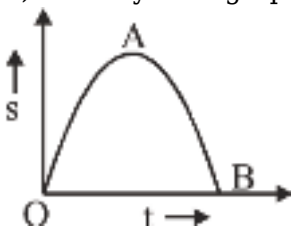
2) A man in a balloon rising vertically with an acceleration of 5 ms^{-2} releases a ball 2 second after the balloon is let go from the ground. The greatest height above the ground reached by the ball is : (g = 10 ms^{-2})

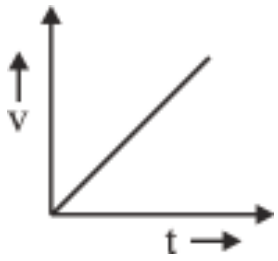
(1) 15 m
(2) 20 m
(3) 10 m
(4) 25 m

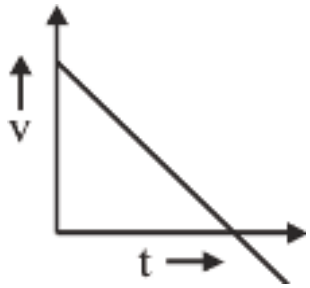
3) The motion of a particle along a straight line is described by equation
$x = 8 + 12t - t^3$
where x is in metre and t in second. The retardation of the particle when its velocity becomes zero, is :-

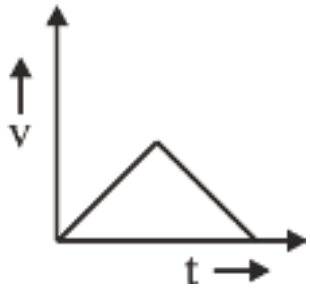
(1) $6ms^{-2}$
(2) $12ms^{-2}$
(3) $24ms^{-2}$
(4) zero

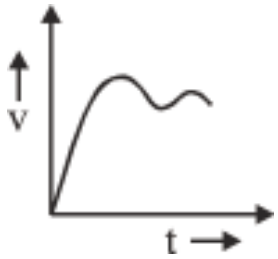
4) Velocity-time graph corresponding to displacement-time graph shown in adjoining figure is :-



(1)


(2) 

(3) 

(4) 

5)

A man walks 30m. towards north, then 20m towards east and in the last $30\sqrt{2}$m towards south-west. The displacement from origin is :-

(1) 10 m towards west

(2) 10m towards east

(3) $60\sqrt{2}$ m towards north west

(4) $60\sqrt{2}$ m towards north east

6) An athlete completes one round of circular track of radius R in 40 seconds. What will be the ratio of distance and magnitude of displacement at the end of 1 minute 40 second?

(1) Zero

(2) $\frac{7\pi}{2}$

(3) $\frac{3\pi}{2}$

(4) $\frac{5\pi}{2}$

7) A particle moves along the curve $y = \frac{x^2}{2}$. Here x varies with time as $x = \frac{t^2}{2}$. Where x and y are measured in metre and t in second. At t = 2s, the velocity of the particle (in ms^{-1}) is :

(1) $2\hat{i} - 4\hat{j}$

(2) $2\hat{i} + 4\hat{j}$

(3) $4\hat{i} + 2\hat{j}$

(4) $4\hat{i} - 2\hat{j}$

8) A car accelerates from rest at constant rate of $2m/s^2$ for some time. Then, it retards at constant rate of 4 m/s^2 and comes to rest. If the total time for which it remains in motion is 3 seconds, what is the total distance travelled ?

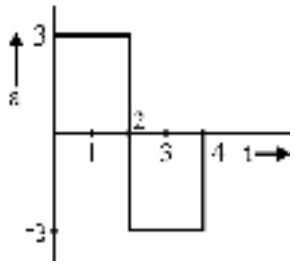
(1) 2 m

(2) 3 m

(3) 4 m

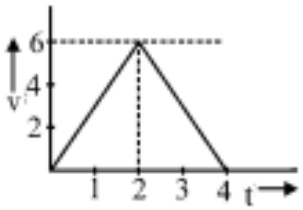
(4) 6 m

9) A particle starts from rest at t = 0 and undergoes an acceleration a in ms^{-2} with time t in second which is as shown :

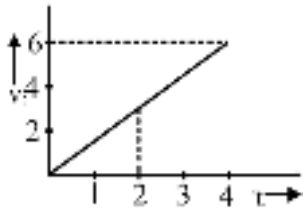


Which one of the following plot represents velocity v in ms^{-1} versus time t in seconds

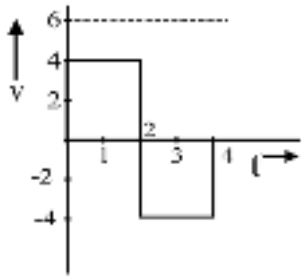
(1)



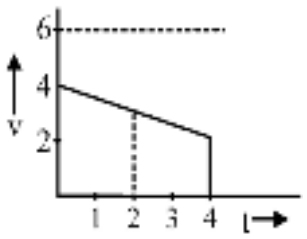
(2)



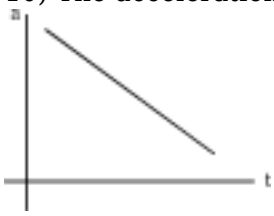
(3)



(4)



10) The acceleration-time graph of a body moving in a straight line is shown here.



Statement-1 : Velocity of the body is continuously decreasing over the time interval shown. and
Statement-2 : Acceleration of the body is continuously decreasing over the time interval shown.

(1) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(2) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(3) Statement-1 is true, statement-2 is false.

(4) Statement-1 is false, statement-2 is true.

11) A ball is thrown upwards. It returns to ground describing a parabolic path. Which of the following remains constant?

(1) Speed of the ball

(2) Kinetic energy of the ball

(3) Vertical component of velocity

(4) Horizontal component of velocity.

12) A boy can throw a stone up to a maximum height of 10 m. The maximum horizontal distance that the boy can throw the same stone up to will be :

(1) $20\sqrt{2}$m

(2) 10 m

(3) $10\sqrt{2}$m

(4) 20 m

13) For two angles of projection of a projectile, $(45° + \theta)$ and $(45° - \theta)$ the horizontal ranges described by the projectile are in the ratio of (if $\theta \leq 45°$) :

(1) 2 : 1

(2) 1 : 2

(3) 1 : 1

(4) 2: 3

14) If the range of a projectile is n times the maximum height of the projectile, then the angle of projection θ is :

(1) $\tan^{-1}(4n)$

(2) $\tan^{-1}\left(\frac{n}{4}\right)$

(3) $\tan^{-1}\left(\frac{4}{n}\right)$

(4) None of these

15) The equation of projectile is $y = 13x - \frac{x^2}{3}$ the horizontal range is :

(1) 16 m

(2) 8 m

(3) 39 m

(4) 3 m

16) A man running at a speed of 5 m/s, the rain drops appear to be falling at an angle of 45° from the vertical. If the rain drops are falling vertically downwards w.r.t. ground, then velocity (w.r.t. ground) in m/s is

(1) 5

(2) $5\sqrt{3}$

(3) $5\sqrt{2}$

(4) 4

17) Rain drops are falling with velocities $(2\hat{i} - 4\hat{j})$ m/sec what should be the velocities of a man so that rain drops hit him with speed 5m/sec.

(1) $-\hat{i}$

(2) $5\hat{i}$

(3) $2\hat{i}$

(4) Both (1) and (2)

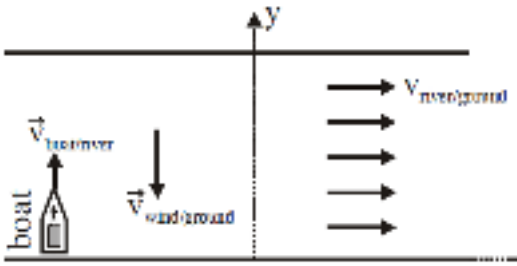
18) A boat takes 2 hours to go 8 km and come back in still water lake. With water velocity of 4km/h,the time taken for going upstream of 8 km and coming back is :

(1) 140 min

(2) 150 min

(3) 160 min

(4) 170 min

19) Figure shows a river flowing towards right. A boat moves perpendicular to river flow. The wind is flowing towards negative y-direction. 

Direction of flutter of flag (relative to an observer on boat) is best represented by which of the following:-

(1) 

(2) 

(3) 

(4) 

20) The speed of a boat is 5 km/h in still water. It crosses a river of width 1 km along the shortest possible path in 15 minutes. The velocity of the river water is :

(1) 1 km/h

(2) 3 km/h

(3) 4 km/h

(4) 5 km/h

21) The wind is blowing in south with velocity of 4 km/h and observer is travelling with velocity of $4\sqrt{3}$ km/h in east. Find out the velocity of wind appeared to observer.

(1) 8 km/hr 30° S of W

(2) 10 km/hr 60° S of W

(3) 8 km/hr 30° E of S

(4) 10 km/hr 30° N of E

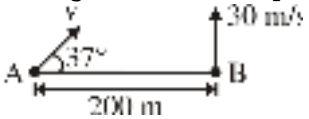
22) A boat moves relative to water with a velocity which is 1/n times the river flow velocity. At what angle to the stream direction must be boat move to minimize drifting ? (n > 1)

(1) $\frac{\pi}{2}$

(2) $\sin^{-1}\left(\frac{1}{n}\right)$

(3) $\pi + \cos^{-1}\left(\frac{1}{n}\right)$

(4) $\pi - \cos^{-1}\left(\frac{1}{n}\right)$

23) Two particles A and B are moving in horizontal plane as shown in figure at t = 0, then time after which A will catch to B will be- 

(1) 4 second

(2) 5 second

(3) 2 second

(4) 8 second

24) A ship is travelling due east at 20 km/h. Find the speed of second ship heading 37° north of east, if it is always due north of the first ship.

(1) 25 km/hr

(2) 12.5 km/hr

(3) 50 km/hr

(4) 33.3 km/hr

25) A passenger in a moving train tosses a coin. If the coin falls behind him, the train must be moving with

(1) an acceleration

(2) a deceleration

(3) a uniform velocity

(4) any of the above

26) Motion of a projectile as observed from other projectile is :-

(1) parabolic

(2) may be parabolic or straight line

(3) straight line

(4) None

27) A man walks for some time 't' with velocity (v) due east. Then he walks for same time 't' with velocity (v) due north. The average velocity of the man is :

(1) 2v

(2) $\sqrt{2}V$

(3) v

(4) $\frac{V}{\sqrt{2}}$

28) During a trip Shyam walks a little faster than Ram, who started off earlier. Shyam observes that he reaches a certain tree, time t_1 later than Ram. Then he catches up with Ram after another lapse of time t_2. The ratio of v_{shyam} to v_{Ram} is

(1) $\frac{t_2}{t_1 + t_2}$

(2) $\frac{t_1 + t_2}{t_2}$

(3) $\frac{t_2}{t_1}$

(4) $\frac{t_1}{t_2}$

29) A train of length 75m long, is to cross a straight bridge of length 425 m. If train is moving with constant velocity of 90 km/hr, then find time taken by it to cross the bridge :-

(1) 10 sec

(2) 5 sec

(3) 20 sec

(4) 17 sec

30) A body starts moving with a velocity $v_0 = 10\ ms^{-1}$. It experiences a retardation equal to $0.2v^2$. Its velocity after 2s is given by-

(1) $+2\ ms^{-1}$

(2) $+4\ ms^{-1}$

(3) $-2\ ms^{-1}$

(4) $+6\ ms^{-1}$

31) The displacement of a body in motion is given by $x = a \sin(\omega t + \theta)$. The time at which the displacement is maximum is (w & q are constants)

(1) $\frac{\theta}{\omega}$

(2) $\left(\frac{\pi}{2\omega} - \frac{\theta}{\omega}\right)$

(3) $\pi / 2\omega$

(4) $\left(\frac{2\pi}{\omega} - \frac{\theta}{\omega}\right)$

32) The displacement 'x' of a particle moving along a straight line at time t is given by $x = a_0 + a_1 t + a_2 t^2$. The acceleration of the particle is :-

(1) a_1

(2) a_2

(3) $2a_2$

(4) $3a_2$

33) A particle is projected upwards from the top of a tower. Treat point of projection as origin and upwards as positive. Consider three point on its path such that at (s represents displacement from origin and v represents velocity)
(i) $s > 0, v > 0$
(ii) $s > 0, v < 0$
(iii) $s < 0, v < 0$
Which of the following correctly indicates the distance travelled by particle since its projection to respective points in appropriate order ?

(1) (i) > (ii) > (iii)

(2) (i) = (ii) < (iii)

(3) (i) < (ii) < (iii)

(4) ordering depends on exact location of points

34) A stone is dropped from a height of 125 m from ground. It is stopped after falling a height of 45 m and then again released. The total time taken by the stone to reach the ground will be :

(1) 6 s

(2) 6.5 s

(3) 7 s

(4) 7.5 s

35) A ball is projected vertically up with a velocity 20 m/s, from top of a tower 25 m. How long it will travel before it hits the ground :-

(1) 6s

(2) 5s

(3) 4s

(4) 12s

36) A body is thrown vertically upwards takes 5 seconds to reach the maximum height. The distnace travelled by the body will be same in :-

(1) 1st and 10th second

(2) 2nd and 8th second

(3) 4th and 6th second

(4) Both (2) & (3)

37) A ball of mass m_1 and another ball of mass m_2 are dropped from equal height. If the time taken by the balls are t_1 and t_2, respectively, then :-

(1) $t_1 = t_2$

(2) $t_1 = 2t_2$

(3) $\frac{t_1}{t_2} = \frac{m_1}{m_2}$

(4) $\frac{t_1}{t_2} = \frac{m_2}{m_1}$

38) A football is kicked into the air vertically upwards with velocity u. The velocity of the ball at the highest point is :-

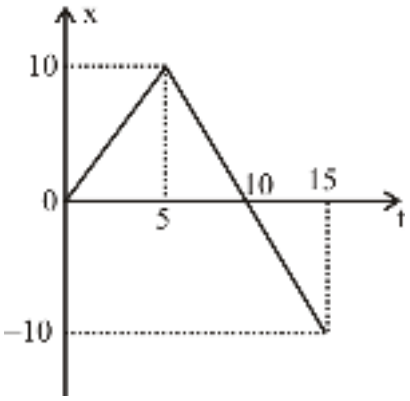
(1) $u\cos\theta$

(2) u

(3) 2u

(4) 0

39) Position-time graph is given, find average velocity in time t = 0 and t = 15 sec is :-



(1) 5 m/s

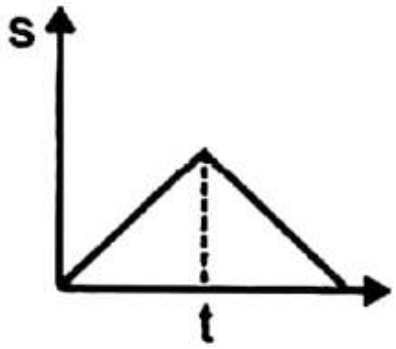
(2) $\frac{5}{3}$ m/s

(3) –2 m/s

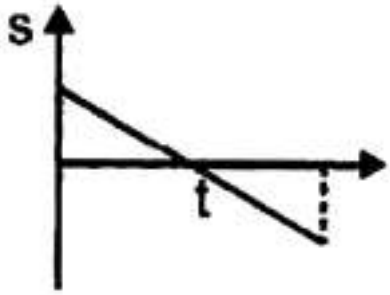
(4) $-\frac{2}{3}$ m/s

40) A body is projected vertically upward from the surface of the earth, its displacement time graph is :

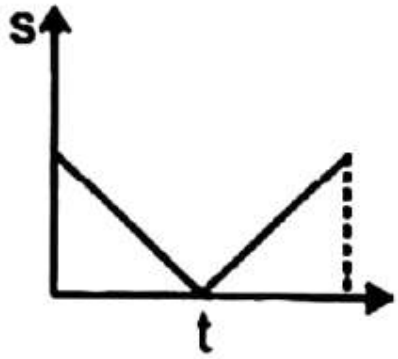
(1)



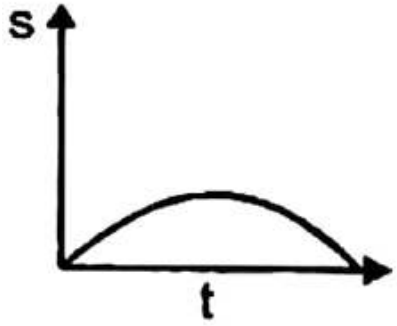
(2)



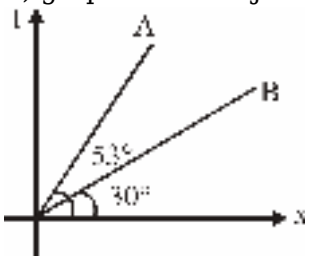
(3)



(4)



41) Time-displacement (t–x) graph of two objects A and B is shown in Fig. The ratio of their speeds (v_A/v_B) is (tan 37° = 3/4):-



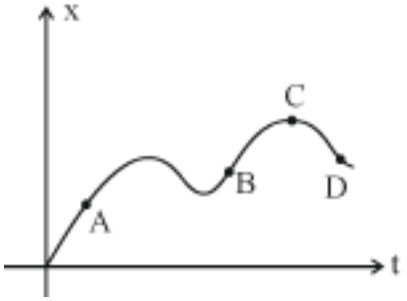
(1) $\frac{\sqrt{3}}{4}$

(2) $\frac{4}{\sqrt{3}}$

(3) $\frac{4}{3\sqrt{3}}$

(4) $\frac{3\sqrt{3}}{4}$

42) From the x – t (displacement-time), the instantaneous velocity is negative at :



(1) B

(2) C

(3) D

(4) A

43) A bird flies for 4 sec with a velocity of |t – 2| m/s in a straight line, where t is in second. It covers a distance of :-

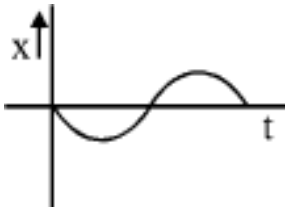
(1) 2m

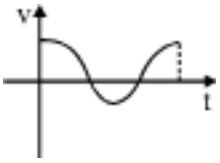
(2) 4m

(3) 6m

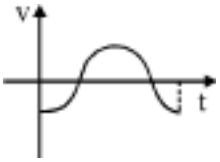
(4) 8m

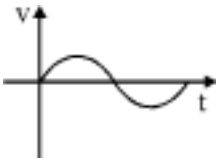
44) If position time graph of a particle is sine curve as shown, what will be its velocity-time graph:-



(1) 

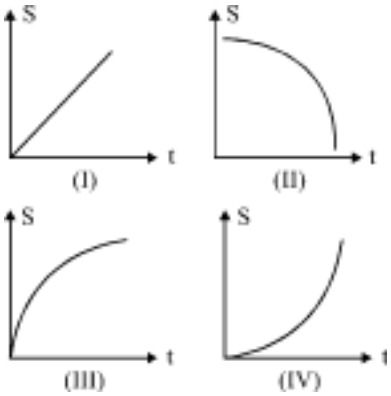
(2) 

(3) 

(4) 

45)

The acceleration will be positive in :-



(1) (I) and (III)

(2) (IV)

(3) (II) and (IV)

(4) None of these

CHEMISTRY

1) Which of the following statement is incorrect ?

(1) During N_2^+ formation, one electron is removed from the bonding molecular orbital of N_2

(2) During O_2^+ formation, one electron is removed from the antibonding molecular orbital of O_2.

(3) During O_2^- formation, one electron is added to the bonding molecular orbital of O_2.

(4) During B_2^- formation, one electron is added to the bonding molecular orbital of B_2.

2) If the magnetic moment of a dioxygen species is 1.73 B.M, it may be :

(1) O_2^- or O_2^+

(2) O_2 or O_2^+

(3) O_2 or O_2^-

(4) O_2, O_2^- or O_2^+

3) Which of the following is not a perfect tetrahedral species?

(1) CCl_4

(2) NH_4^+

(3) $POCl_3$

(4) SiF_4

4) The ion with maximum number of lone pairs on the central atom is-

(1) ClO_2^+

(2) $IC\ell_2^-$

(3) $C\ell O_2^-$

(4) $IC\ell_2^+$

5) Which of the following has all equal bond lengths?

(1) SF_4

(2) IC_3

(3) PCl_3

(4) PCl_5

6) All florine atoms are in same plane in

(1) CHF_3

(2) ClF_3

(3) $XeOF_4$

(4) All of these

7) In which of the following maximum no. of pπ-dπ bonds are present ?

(1) SO_3

(2) XeO_3F_2

(3) NO_2^-

(4) CO_2

8) In which coordinate bond is present ?

(1) NO_2^-

(2) NO_3^-

(3) CO_3^{-2}

(4) All having coordinate bond

9) Which among the following is of highest dipole moment? [NCERT XII-(ii)-294]

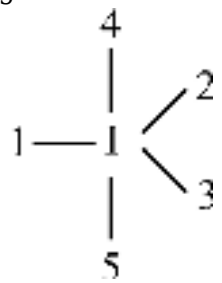
(1) CH_3F

(2) CH_3Br

(3) CH_3CI

(4) CH_3I

10) In order that the diagram below correctly depict the structure of ICl_2^- , at what position should the Cl atoms be drawn ?



(1) 1, 2

(2) 1, 4

(3) 2, 3

(4) 4, 5

11)

In the following reaction, the Cl – P – Cl bond angles :

$$PCl_5 \rightarrow PCl_4^+ + Cl^-$$

(1) All increase

(2) All stay the same

(3) All decrease

(4) Some increase and some decrease

12) Which of the following molecule does not contains bond angle of 90° –

(1) IF_7

(2) $SiCl_4$

(3) XeF_4

(4) PCl_5

13) Among the following pairs, in which the two species are not isostructural :

(1) SiF_4 and SF_4

(2) IO_3^- and XeO_3

(3) BH_4^- and NH_4^+

(4) PF_6^- and SF_6

14) Correct bond angle order is present in :-

(1) $OF_2 < H_2O < OCl_2$

(2) $OCl_2 < H_2O < OF_2$

(3) $H_2O < OF_2 < OCl_2$

(4) $OF_2 < OCl_2 < H_2O$

15) Maximum bond angle is present in :-

(1) H_2O

(2) H_3O^+

(3) NH_4^+

(4) CO_2

16) Number of molecules or ions which has/have any bond angle of 120° is :-
CH_3^-, CH_3^+, NH_3, CO_3^{-2}, BCl_3, PCl_5

(1) 3

(2) 4

(3) 5

(4) 6

17) In which of the following set, the value of bond order will be 2.5:-

(1) O_2^+, NO, NO^{+2}, CN

(2) CN, NO^{+2}, CN^-, F_2

(3) O_2^+, NO^{+2}, O_2^{+2}, CN^-

(4) O_2^{-2}, O_2^-, O_2^+ O_2

18)

Which of the following is polar Molecule :-

(1) XeF_2

(2) XeF_4

(3) N_2O_5

(4) B_2H_6

19) Dipole-Dipole interaction exist between which pair :-

(1) KCl, H_2O

(2) C_6H_6, C_2H_5OH

(3) C_6H_6, CCl_4

(4) Acetone & acetonitrile

20) In which of the following species the hybrid state of the central atom is same ?

(1) SO_2, SO_3

(2) SO_3, SO_4^{2-}

(3) SO_2, SO_3^{2-}

(4) CH_4, HCOOH

21) The strongest carbon-oxygen bond is found in the species :

(1) CO

(2) CO_2

(3) CO_3^{2-}

(4) $CH_3C\overset{\ominus}{O}O$

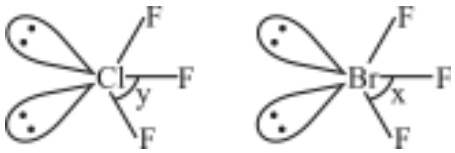
22) s-p mixing present in :-

(1) O_2

(2) F_2

(3) C_2

(4) O_2^{-2}

23) Compare bond angle the following molecules 

(1) x > y

(2) x < y

(3) x = y

(4) None

24) Which molecule has unpaired electron :-

(1) KO_2

(2) BaO_2

(3) Na_2O_2

(4) None

25) Correct order of volatility is:-

(1) HF > HCl > HBr > HI

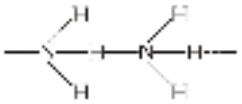
(2) HCl > HBr > HI > HF

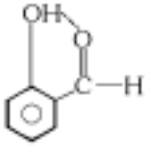
(3) HI > HBr > HCl > HF

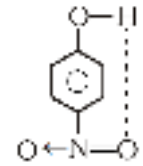
(4) HBr < HCl < HI < HF

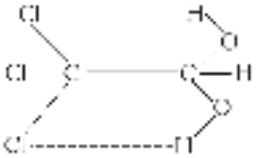
26)

In which of the following molecule, the shown hydrogen bond is not possible:-

(1) 

(2) 

(3) 

(4) 

27) Intermolecular hydrogen bonds are not present in:-

(1) CH_3CH_2OH

(2) CH_3COOH

(3) $C_2H_5NH_2$

(4) CH_3OCH_3

28) The compound containing co-ordinate bond is :

(1) H_2SO_4

(2) O_3

(3) SO_3

(4) All of these

29)

Consider the following table :-

Steric no. (σ bp + □.p.)	□.p. on central atom	Shape of molecule
5	a	Linear
b	1	See-Saw
6	c	Square Planar
d	2	T-Shaped

The value of a + b + c + d is :-

(1) 10

(2) 9

(3) 15

(4) 8

30) Which of the following has sp^3d^2 hybridisation :-

(1) $XeOF_4$

(2) I_3^-

(3) SF_4

(4) XeF_5^-

31) C_7H_9N has how many isomeric forms that contain a benzene ring :-

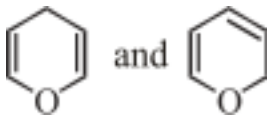
(1) 4

(2) 5

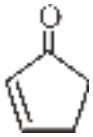
(3) 3

(4) 6

32) Which of the following pairs of structure do not represent isomer :-

(1)  and

(2)  and 

(3)  and 

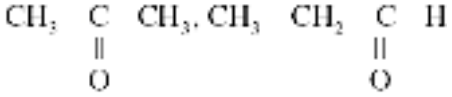
(4)  and 

33) Which are metamers :

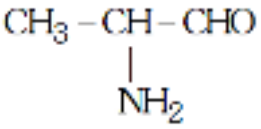
(1) $CH_3–O–CH_2CH_2CH_3$, $CH_3–CH_2–O–CH_2–CH_3$

(2) C_2H_5 –O–C_2H_5, $CH_3CH_2CH_2CH_2OH$

(3) $CH_3–O–C_2H_5$, $CH_3–CH_2–O–CH_3$

(4) 

34) (I) $CH_3 – CH_2 – NH – CHO$;

(II) 

Which type of isomerism is observed between I and II.

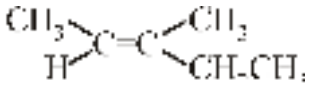
(1) Chain isomers

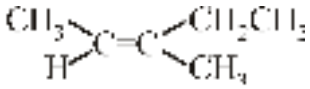
(2) Position isomers

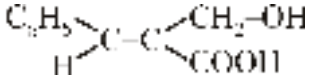
(3) Functional isomers

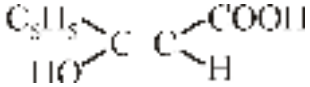
(4) Metamers

35) The Z-isomer among the following is :-

(1) 

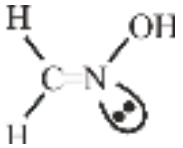
(2) 

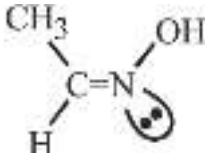
(3) 

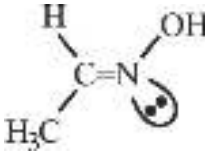
(4) 

36)

Which of the following is syn aldoxime :-

(1) 

(2) 

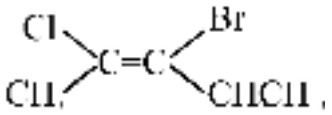
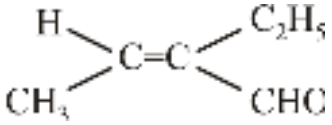
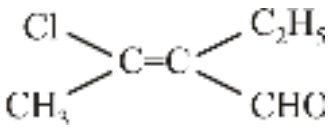
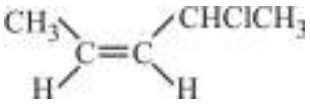
(3) 

(4) All

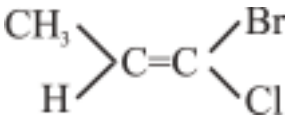
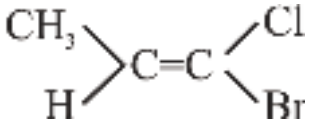
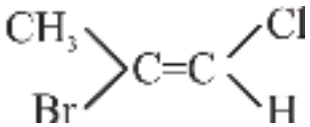
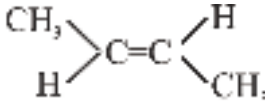
37) Geometrical isomerism result because the molecule :-

(1) Rotates plane polarized light
(2) Has a plane at symmetry
(3) Has a centre of symmetry
(4) Hindered rotation

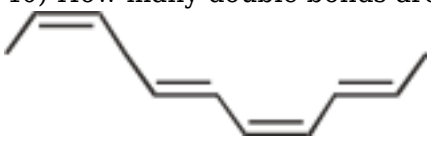
38) Which among the following has E configuration?

(1) 
(2) 
(3) 
(4) 

39) Which of the following is a Z-isomer :-

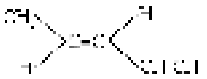
(1) 
(2) 
(3) 
(4) 

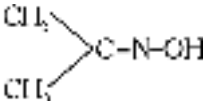
40) How many double bonds are having cis arrangement in following example :

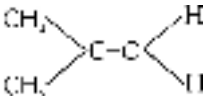


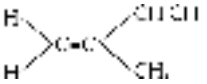
(1) One
(2) Two
(3) Three
(4) Four

41) Which exhibits Geometrical isomerism :–

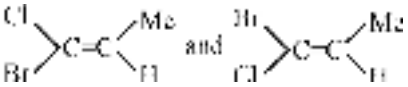
(1) 

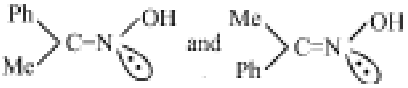
(2) 

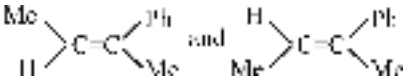
(3) 

(4) 

42) Which is a pair of geometrical isomers :-

(1) 

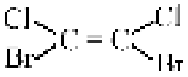
(2) 

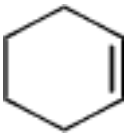
(3) 

(4) All of these

43) Which one show G.I. & Non-planer ?

(1) 

(2) 

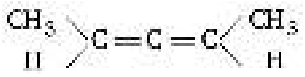
(3) 

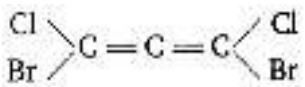
(4) 

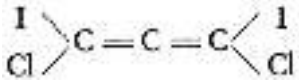
44) Compound which can show G.I.?

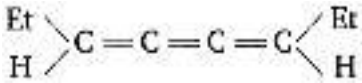
(1) 2-phenyl prop-1-ene
(2) 1,1-dichloro prop-1-ene
(3) 4-phenyl but-1-ene
(4) 1-phenyl but-2-ene

45) Which of the following will show G.I.?

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) Mesophyll cell, Chloroplast, Light, Atmospheric CO_2 concentration, Temperature, Water, Amount of chlorophyll.
How many of the above factors belongs to external factor of plants which affect photosynthesis ?

(1) 5

(2) 6

(3) 4

(4) 7

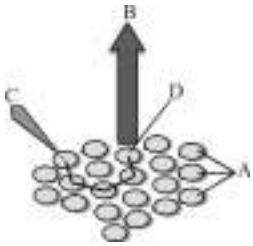
2) Photosynthesis is a/an :-
(a) Endergonic process
(b) Anabolic process
(c) Physico chemical process

(1) Only a, b

(2) Only c

(3) All a, b, c

(4) Only b

3) Choose the correct match for A to D:- 

(1) A- Photons, B- Reaction centre, C- Primary acceptor, D- accessory pigments

(2) A- Antenna molecules, B-Primary acceptor , C- Photons, D-Reaction centre

(3) A-Primary acceptor, B-Photons, C-Reaction centre, D- Antenna molecules

(4) A-Antenna molecules, B- Reaction centre C-photons, D- Primary Acceptor

4) **Statement A :** Biosynthetic reaction is independent of direct presence of light.
Statement B: Biosynthetic process continues for some time after the light becomes unavailable and then stops.

(1) Both A & B are correct

(2) A is correct and B incorrect

(3) A is incorrect and B is correct

(4) A & B are incorrect

5) Read the following five statements (A - E) :-
(a) Movement of electrons in Z-scheme is always down hill
(b) Primary electron acceptor of z-scheme is located toward outer side of thylakoid membrane
(c) Antenna complex is composed of only carotenoids
(d) NADP reductase enzyme is located toward outer side of thylakoid membrane
(e) Water splitting complex is located on stromal side of thylakoid membrane.
How many of the above statements are correct ?

(1) Four
(2) One
(3) Two
(4) Three

6) Which of the following component is not utilised in the type of photophosphorylation in which there is no formation of $NADPH.H^+$?

(1) Reaction centre with absorption peak of 700 nm wavelength of light.
(2) Reaction centre which causes the release of O_2 during light reaction.
(3) Cytochrome containing electron transport system to pass the electrons.
(4) An iron-sulphur protein as primary electron acceptor.

7) Consider the following four statements (a–d) and select which includes the correct ones only :-
(a) The splitting of water is associated with the PS–II.
(b) During Photophosphorylation ATP are form in down hill process.
(c) When only PS–II is functional the electron is circulated with in the photosystem.
(d) In noncyclic phosphorilation only PS–II is involves.

(1) Statement a, b, c
(2) Statement a, b
(3) Statement a, b, c, d
(4) Statement a, c, d

8) Find the **incorrect** statement :-

(1) Breakdown of H^+ gradient across the thelakoid membrane can form ATP
(2) In 'Z' scheme flow of e^- is uphill and down hill both
(3) Only cyclic photophosphorylation will happens if wavelength of light is more than 680 nm upto 700 nm of PAR
(4) Productivity of C_3 plant is more than C_4 plant with respect to photosynthesis

9) Identify the correct match :-

	A(Condition)	**B(Limiting factor)**
(1)	Morning time	CO_2
(2)	Cloudy sky	CO_2
(3)	Dense forest	Light intensity

(4)	Clear sky	Light intensity

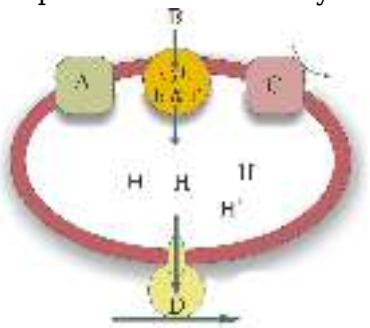
(1) 1

(2) 2

(3) 3

(4) 4

10) Given below is the diagrammatic sketch of a certain phenomenon. Identify the parts labelled A,



B, C and D and select the right option about them :-

	Part A	Part B	Part C	Part D
(1)	Site of photolysis of H_2O	Cyt. a-a_3	Site of $NADPH.H^+$ formation	F_0 - F_1 complex
(2)	PS-I	H^+	PS-II	Site of ATP synthesis
(3)	Site of $NADPH.H^+$ formation	Site of splitting of H_2O	PS-I	F_0 - F_1 complex
(4)	PS-II	H^+	PS-I	Site of ATP synthesis

(1) 1

(2) 2

(3) 3

(4) 4

11)

The calvin cycle completes in three steps :-
(A) Utilization of hydrogen given by $NADPH{\cdot}H^+$.
(B) Regeneration of primary CO_2 acceptor.
(C) Reaction of RuBP with CO_2.

Identify the correct sequence during the process:-

(1) A → C → B

(2) C → A → B

(3) B → A → C

(4) C → B → A

12) How many of the following represents the correct match of discoverer & discoveries ?

(A)	Jan Ingenhousz	O_2 evolved from H_2O not from CO_2
(B)	Engelmann	Action spectrum of photosynthesis
(C)	Julius von Sachs	Relation among photosynthesis, chloroplast & starch
(D)	Van Niel	Bell-Jar experiment

(1) One

(2) Two

(3) Three

(4) Four

13) Read the following statements :-
(i) In Engelmann's experiment the aerobic bacteria accumulated mainly in the region of green light of the split spectrum
(ii) Action spectrum of photosynthesis corresponds closely to absorption spectra of chlorophyll-a showing that chlorophyll-a is the chief photosynthetic pigment
(iii) Some organisms do not release O_2 during photosynthesis
(iv) In low light intensitiy chloroplast aligned perpendicular to the incident light.
Which of the above statements are correct ?

(1) (i), (ii) and (iii)

(2) (i) only

(3) (ii), (iii) and (iv)

(4) (iii) and (iv)

14) **Statement-I :** Photochemical phase of photosynthesis includes light absorption, water splitting, oxygen release and formation of high energy chemical intermediates ATP & NADPH.
Statement-II : Photosystem-I and photosystem-II are named in the sequence of their function during the light reaction.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Statement-I & Statement-II both are incorrect.

15) **Statement-I :** Light is a rare limiting factor for photosynthesis in nature
Statement-II : C_4 plants show saturation at 360 ppm of atmospheric CO_2 concentration.

(1) Statement I and II both are correct.

(2) Statement I and II both are incorrect.

(3) Only Statement I is correct.

(4) Only Statement II is correct.

16) The first CO_2 acceptor and first stable product in C_4 and C_3 cycle respectively are:

(1) 3-C, 4-C ; 5-C, 3-C
(2) 5-C, 3-C ; 2-C, 4-C
(3) 5-C, 3-C ; 3-C, 4-C
(4) 2-C, 3-C ; 4-C, 3-C

17) Find the incorrect statement :

(1) In bundle sheath cells of C_4 plants, first product of CO_2 fixation is 5C compound.
(2) In photorespiration both chloroplast and peroxisome utilize O_2
(3) PEP carboxylase plays important role in synthesis of C_4 acid in mesophyll cells of C_4 plants.
(4) ATP requirement of C_4 plants is higher as compared to C_3 plants to produce one molecule of sugar.

18) Which of the following combinations is required for chemiosmosis?

(1) Membrane, Proton Pump, NADP Synthase
(2) Proton Pump, Electron gradient
(3) Membrane, Proton pump, Proton gradient, ATP Synthase
(4) Proton Pump, Electron gradient, NADP Synthase

19) **Assertion :** C_4-pathway is more efficient than C_3 -pathway.
Reason : Photorespiration is supressed in C_4-plants.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(3) **Assertion** is true but **Reason** is false.
(4) Both **Assertion** and **Reason** are false.

20) How many components listed below are part of cyclic ETS?

P_{700}, P_{680}, NADP reductase, PS I, water splitting complex, PS II

(1) Two
(2) Three
(3) Five
(4) Four

21) Given below are two statements :
Statement-I :- Photorespiration is the process which create an important difference between C_3 and C_4 plant.
Statement-II :- In the photorespiratory pathway there is neither synthesis of sugars, nor of ATP and NADPH
In the light of the above statements, choose the correct answer from the options given below :

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

22) Match the following columns :-

	Pigment		**Colour**
(A)	Chlorophyll b	I.	Yellow
(B)	Carotene	II.	Orange
(C)	Xanthophyll	III.	Blue green
(D)	Chlorophyll a	IV.	Yellow green

(1) A- I, B-II ,C -III, D-IV

(2) A-IV,B-II, C- I, D- III

(3) A-II, B-III,C-I,D-IV

(4) A-III,B-IV,C-II,D-I

23) Select correct statement –

(1) C_4 plants show CO_2 saturation at 450 $\mu\ell L^{-1}$.

(2) C_3 plants show CO_2 saturation at 360 $\mu\ell L^{-1}$.

(3) C_3 plants show higher productivity at higher CO_2 concentration.

(4) C_4 plants are benefitted if they grown in CO_2 enriched environment.

24) How many photons required for the evolution of one molecule of oxygen in light reaction?

(1) eight photons

(2) four photons

(3) sixteen photons

(4) not specific

25) **Statement-I :** In C_4 plants the CO_2 acceptor in Calvin cycle is present in bundle sheath cells.
Statement-II : C_4 plants have both RuBisCO and PEPcase enzymes.

(1) Only statement-I is correct

(2) Only statement-II is correct

(3) Both statement-I and II are correct

(4) Both statement-I and II are incorrect

26) **Statement-I :** Second step of calvin cycle (reduction) involves utilization of 2 molecules of ATP for reduction and 2 molecules of NADPH for phosphorylation per CO_2 molecule fixed.
Statement-II : The regeneration steps require one ATP for phosphorylation to form RuBP.

(1) Only Statement I is correct

(2) Only Statement II is correct

(3) Statement I and II both are correct

(4) Statement I and II both are incorrect



27) What happens next in given figure of Priestley's experiment ?

(1) Mouse will suffocate

(2) Burning candle gets extinguish

(3) Both (1) and (2)

(4) Mouse stays alive and the candle continue to burn

28) Select the correct statements.
A. In high light intensity chloroplasts will be aligned with their flat surfaces parallel to the lateral walls.
B. In low light intensity chloroplasts will be aligned perpendicular to the incident light.
C. There is a clear division of labour within the chloroplast.
D. O_2 evolved by the purple and green bacteria comes from water, not from carbon dioxide.
Choose the correct answer from the option given bellow :

(1) A and B only

(2) A and D only

(3) A, B and C only

(4) A, C and D only

29) Which of the following is correct for photorespiration?

(1) Synthesis of ATP with release of CO_2

(2) Synthesis of sugar and NADPH both

(3) Release of CO_2 and utilisation of ATP

(4) Synthesis of NADPH but not of sugar

30) Which of the following is correct for photorespiration ?

(1) RuBP + $O_2 \rightarrow$ 2, Phosphoglycerate

(2) RuBP + $CO_2 \rightarrow$ 2, Phosphoglycerate

(3) RuBP + $CO_2 \rightarrow$ Phosphoglycerate + Phosphoglycolate

(4) RuBP + $O_2 \rightarrow$ Phosphoglycerate + Phosphoglycolate

31) Which of the following factors favour the oxygenase activity of RuBisCO ?

(1) $O_2\downarrow, CO_2\uparrow$, Light intensity$\downarrow$, temperature$\uparrow$

(2) $CO_2\uparrow, O_2\downarrow$, Light intensity$\uparrow$, temperature$\downarrow$

(3) $O_2\downarrow, CO_2\downarrow$, Light intensity$\downarrow$, temperature$\downarrow$

(4) $CO_2\downarrow, O_2\uparrow$, Light intensity$\uparrow$, temperature$\uparrow$

32) For the fixation of 12 molecules of CO_2 in Rice and Maize, how many molecules of ATP are required respectively ?

(1) 18 and 30

(2) 36 and 60

(3) 3 and 5

(4) 18 and 60

33) In Hatch & Slack pathway decarboxylation of C4 acid and regeneration of phosophoenol pyruvate occurs respectively in :-

(1) mesophyll cell and bundle sheath cell

(2) bundle sheath cell and mesophyll cell

(3) bundle sheath cell and bundle sheath cell

(4) mesophyll cell and mesophyll cell

34) Match the following, and choose the correct option :-

1	Non cyclic ETS	a	Grana and Stroma thylakoid
2	Cyclic ETS	b	Grana thylakoid
3	Chemiosmotic theory	c	Stroma
4	Dark reaction	d	Formation of ATP

(1) 1-a, 2-b, 3-c, 4-d

(2) 1-b, 2-a, 3-d, 4-c

(3) 1-d, 2-c, 3-b, 4-a

(4) 1-a, 2-b, 3-d, 4-c

35) ***Assertion :-*** There is linear relationship between incident light and CO_2 fixation rates at low light intensities.
Reason :- In this condition CO_2 become the limiting factor.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Assertion is false but Reason is true.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

36) **Assertion :** Protons produced by the splitting of water accumulate within the lumen of

thylakoid.
Reason : Splitting of water molecule take place on inner side of the membrane of thylakoid.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

37) **Assertion :** Single chlorophyll-a molecule form the reaction centre.
Reason : The LHC are made up of hundreds of pigment molecules.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Assertion is False but the Reason is True.

38) **Assertion :-** Photosynthesis is a physico-chemical process.
Reason :- In photosynthesis light energy convert into chemical energy.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

39) Given below are two statements : One is labelled as **Assertion-A** and the other is labelled as **Reason-R**.
Assertion (A) : In C_4 plants photorespiration does not occur.
Reason (R) : C_4 plants have mechanism that increases the concentration of CO_2 at the site of enzyme RuBisCO
In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both A and R are true but R is NOT the correct explanation of A.
(2) A is true but R is false.
(3) A is false but R is true.
(4) Both A and R are true and R is the correct explanation of A.

40) **Assertion (A) :-** To make one molecule of glucose 6 turns of calvin cycle are required.
Reason (R) :- In each turn of calvin cycle there is fixation of one CO_2 only, while $6CO_2$ are required for each glucose.
In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
(2) **A** is true but **R** is false.
(3) **A** is false but **R** is true.
(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

41) **Assertion :-** There is not a complete one- to-one overlap between the absorption spectrum of

chlorophyll a and the action spectrum of photosynthesis.
Reason : Most of the photosynthesis takes place in the blue and red regions of the spectrum but some photosynthesis does take place at the other wavelengths of the visible spectrum.

(1) Assertion and reason both are true but reason is not the correct explanation of the assertion
(2) only Assertion is true
(3) Both assertion and reason are false
(4) Assertion and reason both are true and reason is the correct explanation of the assertion

42) The Z scheme is named so because-

(1) It was discovered by a scientist with 'Z' as initial letter of name
(2) The carriers of ETS present in thylakoid membrane are in 'Z' shape.
(3) It forms 'Z' shape when the carriers of ETS are arranged in sequence of redox potential scale.
(4) Both (1) and (3)

43) How many additional ATPs are required in C_4 plants as compared to C_3 plants to synthesize 2 molecules of hexose sugar ?

(1) 12 ATP
(2) 24 ATP
(3) 36 ATP
(4) 48 ATP

44) Assertion:– Dark reactions are called biosynthetic phase of photosynthesis.
Reason:– Dark reactions do not directly depend on the presence of light but are dependent on the products of the light reaction, i.e., ATP and NADPH.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not correct explanation of (A).
(3) Both (A) is true but (R) is false
(4) Both (A) and (R) are false.

45) (A) The F_1 head piece is a transmembrane protein complex which contain the site of ATP synthesis.
(B) F_0 is an integral membrane protein complex which forms the channel through which proton cross the outer membrane of mitochondria

(1) Only A is correct
(2) Only B is correct
(3) Both A & B are correct
(4) Both A & B are incorrect

46) Nerve cell do not divide beacuse they do not possess :-

(1) Nucleus
(2) Golgi body

(3) Mitochondria
(4) Centriole

47) Schwann cells are found on :-

(1) Myelinated nerve fibres
(2) Unmyelinated nerve fibres
(3) Cell body of neuron
(4) Both (1) and (2)

48) Synaptic vesicle occurs in :-

(1) Presynaptic neuron
(2) Post-synaptic neuron
(3) Synaptic cleft
(4) None of the above

49) The forebrain consists of :-

(1) Cerebellum, thalamus and hypothalamus
(2) Pons, cerebrum and medulla.
(3) Amygdala, hippocampus and brain stem
(4) Thalamus, hypothalamus and cerebrum

50) Association areas is-

(1) Sensory in function
(2) Motor in function
(3) Neither clearly sensory nor motor in function
(4) Neurosecretory in function

51) Group of nerve fibers outside CNS is called as-

(1) Ganglia
(2) Nerve
(3) Tract
(4) Nuclei

52) Which of the following pair is not matched correctly:-

(1) Limbic system ⇒ forebrain
(2) Brain stem ⇒ Only mid brain
(3) Association area ⇒ cerebral cortex
(4) Centres of respiration ⇒ Hind brain

53) Area responsible for analysis of taste is present in which lobe :-

(1) Occipital
(2) Parietal
(3) Frontal
(4) Temporal

54) Cerebral hemispheres are connected by a tract of nerve fibres called as :-

(1) Corpus albicans
(2) Thalamus
(3) Corpus callosum
(4) (2) & (3) both

55) When a neuron is not conducting any impulse i.e. resting, the axonal membrance is -

(1) Comparatively more permeable to K^+ and impermeable (nearly impermeable) to Na^+
(2) Impermeable to negatively charged proteins present in the axoplasm
(3) (1) & (2) Both
(4) More permeable to Na^+ ions than K^+ ion

56) **Assertion :-** Human's brain having highly folded structure in gray matter as gyri & sulci.
Reason :- Gyri & sulci are also present in cerebellum.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the assertion.
(2) Both Assertion & Reason are true but Reason is not a correct explanation of the assertion.
(3) Assertion is true but the Reason is false.
(4) Both Assertion and Reason are false

57) "Jumping of the action potential" at the nodes of ranvier is known as :-

(1) Saltatory conduction
(2) Neuro transmission
(3) Recovery phase
(4) Active phase

58) Outer most covering of brain is called :-

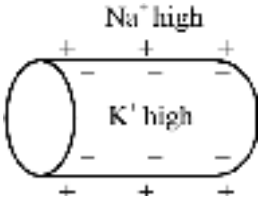
(1) Choroid
(2) Duramater
(3) Piamater
(4) Arachnoid

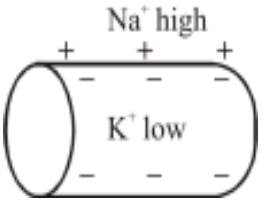
59) During depolarization of the neuronal membrane-

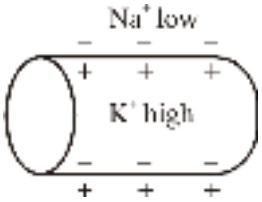
(1) Na^+ ions rapidly move inside the cell
(2) Na^+ ions rapidly move outside the cell
(3) K^+ ions rapidly move outside the cell

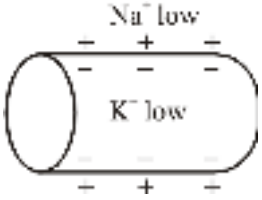
(4) K^+ ions rapidly move inside the cell

60) Which of the following diagram illustrates the distribution of Na^+ and K^+ ions in a section of non-myelinated axon which is at resting potential?

(1) 

(2) 

(3) 

(4) 

61) Hypothalamus is the ___(A)____ part of____(B)___ ?

	A	B
(1)	Dorsal	Diencephalon
(2)	Basal	Diencephalon
(3)	Basal	Mid bain
(4)	Dorsal	Medulla oblongata

(1) 1

(2) 2

(3) 3

(4) 4

62) Transmission of an impulse across ___A___ synapse is very similar to impulse conduction along a single ___B___ . Choose the correct options for A and B

	A	B
(1)	Chemical	Axon
(2)	Electrical	Cyton
(3)	Chemical	Cyton

(4)	Electrical	Axon

(1) 1

(2) 2

(3) 3

(4) 4

63) Read the following flow regarding to voltage :-

$-70\ mv \rightarrow -55mv \rightarrow 0mv \rightarrow +10\ mv$

(A) (B) (C) (D)

at which potential Na^+–K^+ pump is closed :-

(1) A

(2) A & B

(3) B,C & D

(4) A & D

64) Nissl's granules are found in ..... and helps in ..... :-

(1) Cyton only, lipid synthesis respectively

(2) Cyton and Axon, Protein synthesis respectively

(3) Dendron and Axon, Protein synthesis respectively

(4) Dendron and Cyton, Protein synthesis respectively

65) Match the column I and II, and choose the correct option :-

Column -I		Column-II	
a	Unipolar Neuron	1	Cerebral cortex
b	Bipolar Neuron	2	Embryonic stage
c	Multipolar neuron	3	Retina

(1) a-1, b-2, c-3

(2) a-2, b-3, c-1

(3) a-3, b-1, c-2

(4) a-2, b-1, c-3

66) During depolarisation of nerve :-

(1) Na^+ VGC – Close, K^+ VGC – open

(2) Na^+ VGC – Open, Na^+-K^+ pump – on

(3) K^+ VGC – Close, Na^+-K^+ pump – off

(4) Na^+ VGC – Open, K^+ leak channel close

67) Fibres which transmit impulses towards the cell body called as :-

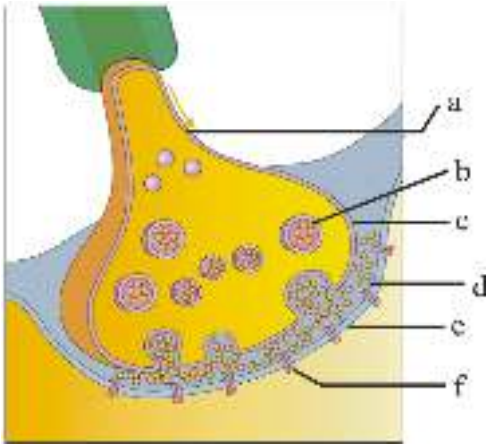
(1) Axon terminal

(2) Axon

(3) Dendrites

(4) Axon hillock

68) Recognise the figure and find out correct matching :



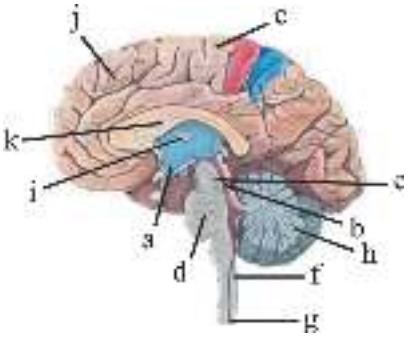
(1) a-Axon, b-synaptic cleft, c-Pre synaptic-membrane, d-synaptic vesicle, e-receptor, f-post synaptic membrane

(2) a-Axon terminal, b-synaptic vesicle, c-Pre synaptic-membrane, d-synaptic cleft, e-post synaptic membrane, f Neurotransmitters

(3) a-synaptic vesicles, b-Axon terminal, c-post synaptic membrane, d-Neurotransmitters, e-Pre synaptic-membrane, f-receptor

(4) a-Axon terminal, b-synaptic vesicles, c-Pre synaptic-membrane, d-synaptic cleft, e-post synaptic membrane, f-receptor

69) Coordinating centre for sensory and motor signaling is:



(1) e

(2) f

(3) d

(4) i

70) The space between the two adjoining neurons where chemical transmitter is released, known as :

(1) Synaptic vesicle

(2) Node of Ranvier

(3) Synaptic cleft

(4) Telodendria

71) Brain stem is made up of -

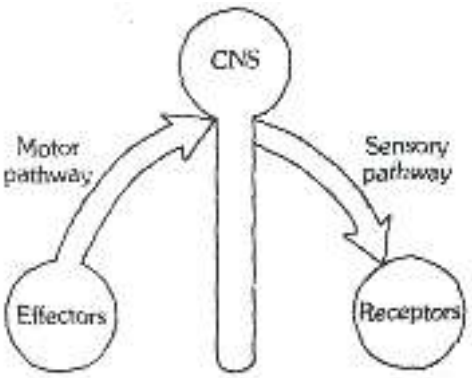
(1) Mid brain and cerebrum
(2) Mid brain, pons and spinal cord.
(3) Pons, medulla oblongata and spinal cord.
(4) Mid brain, pons and medulla oblongata.

72) Na^+ - K^+ pump in axolemma transports :-

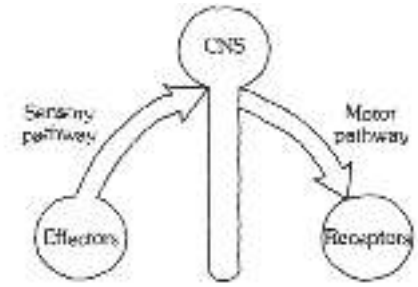
(1) 3 Na^+ out and $2K^+$ in
(2) 3 Na^+ in and $2K^+$ out
(3) 2 Na^+ out and $3K^+$ in
(4) 2 Na^+ in and $3K^+$ out

73) Choose the correct diagram which represent the flow of information through the nervous system:-

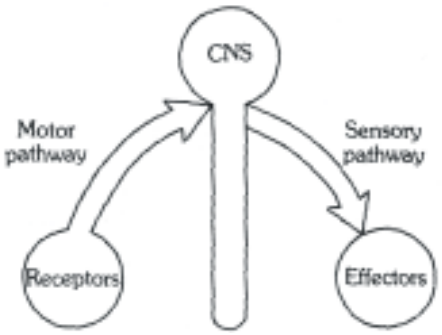
(1)

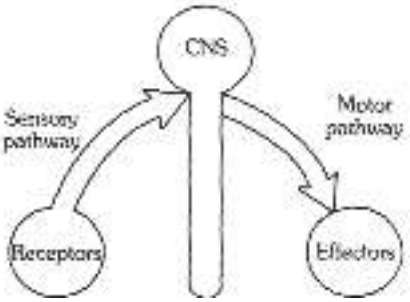


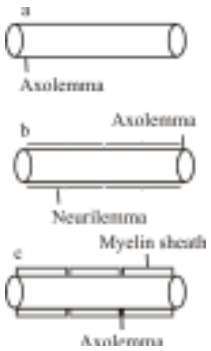
(2)



(3)



(4) 



74) In which of the following, speed of conduction is the fastest ?

(1) a

(2) b

(3) c

(4) a, b

75) Find out the correct match :
(a) Limbic system = regulation of sexual behaviour with hypothalamus.
(b) Thalamus = Co-ordinates sensory and motor signals.
(c) Association = Intersensory association, memory and communication.
(d) Pons = Consist fibre tracts that inter connect different brain regions.

(1) only a, b, c correct

(2) only a, c correct

(3) only b, c, d correct

(4) All correct

76) In resting condition the outer surface of the axonal membrane possesses a __(A)__ while its inner surface becoms __(B)__ and therefore is __(C)__ :-

	(A)	(B)	(C)
(1)	Positive charge	Negative charge	Depolarised
(2)	Positive charge	Negative charge	Polarised
(3)	Negative charge	Positive charge	Polarised
(4)	Positive charge	Negative charge	Hyperpolarise

(1) 1

(2) 2

(3) 3

(4) 4

77) The cerebral cortex is referred to as the __(A)__ due to its __(B)__ appearance the neuron __(C)__

are concentrated here giving the colour :-

	(A)	(B)	(C)
(1)	White mater	White	Cell bodies
(2)	Grey mater	Grey	Axon
(3)	White mater	White	Axon
(4)	Grey mater	Grey	Cell bodies

(1) 1

(2) 2

(3) 3

(4) 4

78) Cerebral aqueduct passes through :-

(1) Hind brain

(2) Fore brain

(3) Midbrain

(4) Spinal cord

79) The part of the peripheral nervous system that comprises the whole complex of nerves, fibres, ganglia and plexuses by which impulses travel from the central nervous system to the viscera and from viscera to central nervous system is called :

(1) Autonomic nervous system

(2) Sympathetic nervous system

(3) Parasympathetic nervous system

(4) Visceral nervous system

80) Parasympathetic nervous system affects :-

(1) Pupillary dilation

(2) Stimulation of secretion of sweat

(3) Stimulation of salivary secretion

(4) Ejaculation

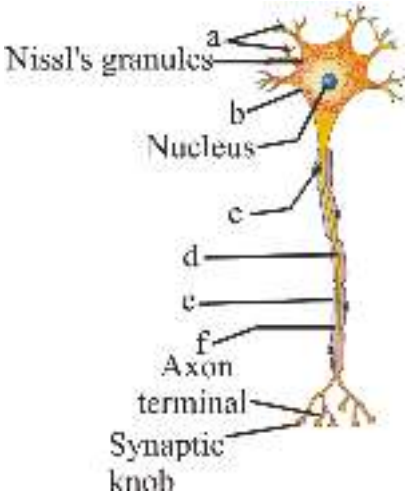
81) Read the following statements carefully.
(i) Axon is the longest cell process of cyton.
(ii) Nissl's granules are abundant in the axoplasm.
(iii) The axon hillock is the nephron's trigger zone.
(iv) Axon transmit nerve impulses towards it's cell body
How many of the above statements are not incorrect?

(1) Three

(2) Two

(3) One

(4) Four



82) Recognise the figure and find out the correct matching :-

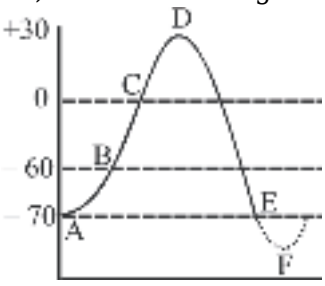
(1) a-axons, b-cell body, c-myelin sheath,
d-dendrites, e-node of Ranvier, f-Schwann cell

(2) a-dendrites, b-axon, c-Schwann cell,
d-cell body, e-myelin sheath, f-node of Ranvier

(3) a-dendrites, b-cell body, c-Schwann cell,
d-axon, e-myelin sheath, f-Node of Ranvier

(4) a- synaptic knobs, b-cell body, c-myelin sheath,
d-axon, e-Schwann cell, f-Node of Ranvier

83) Look at the diagram given below carefuly and give the answer of following questions :



Complete opening of Na^+ V.G.C at :-

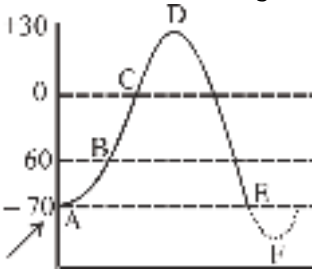
(1) A

(2) B

(3) C

(4) D

84) Look at the diagram given below carefuly and give the answer of following questions :



K^+ V.G.C get closed at:-

(1) C

(2) D

(3) E

(4) F

85) **Assertion :** Neurons are excitable cells.
Reason : Neurons are found abundantly throughout the body.

(1) Both assertion and reason are true, reason is correct explanation of assertion
(2) Both assertion and reason are true, reason is not the correct explanation of assertion
(3) Assertion is true, reason is false
(4) Both assertion and reason are false

86) Match the following parts of human brain with the structures associated with them and choose the correct option :-

Parts of Brain		**Structure**	
(i)	Forebrain	(a)	Corpora quadrigemina
(ii)	Midbrain	(b)	Thalamus
(iii)	Hindbrain	(c)	Medulla

(1) (i)-b (ii)-c (iii)-a
(2) (i)-a (ii)-b (iii)-c
(3) (i)-b (ii)-a (iii)-c
(4) (i)-c (ii)-b (iii)-a

87) Across synapse following steps of transmission of nerve impulse arrange in sequence :-
(A) Opening of specific ions channels allows the entry of ions a new excitatory or inhibitory potential are generated
(B) Arrival of action potential at axon terminal
(C) Synaptic vescicle-fuses with the synaptic membrane & neurotransmitter releases in synaptic cleft by influx of Ca+2 ions
(D) Neurotransmitter bind to receptor on post synaptic membrane

(1) (B) → (C) → (D) → (A)
(2) (D) → (C) → (B) → (A)
(3) (B) → (A) → (C) → (D)
(4) (D) → (B) → (C) → (A)

88) ***Assertion*** :- Schwann cells are present in myelinated & unmyelinated axon.
Reason :- Schwann cell has functions of myelinogenesis in CNS.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

89) **Assertion :** The binding of Neurotransmitter on post synaptic membrane opens ion channels allowing the entry of ions in post synaptic neuron. **Reason :** Entry of ion can generate a new potential may be either excitatory on inhibitory on post synaptic neuron.

(1) Assertion and Reason both are correct and Reason is the correct explaination of Assertion

(2) Assertion and Reason both correct but Reason is not the correct explaination of Assertion

(3) Assertion is correct but Reason is incorrect

(4) Both Assertion & Reason are incorrect

90) **Assertion (A) :-** CNS is main site of information processing and control.
Reason (R) :- PNS comprises of all the neurons of body associated with CNS.

(1) Both A and R are true and R is correct explanation A

(2) Both A and R are true and R is not correct explanation A

(3) Only A is True

(4) Both A and R are false

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	1	2	2	1	4	2	4	1	4	4	4	3	3	3	1	4	3	2	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	4	2	1	1	3	4	2	3	1	2	3	3	3	2	1	1	4	4	4
Q.	41	42	43	44	45															
A.	1	3	2	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	1	3	2	3	4	2	2	3	4	4	2	1	1	4	2	1	3	4	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	3	2	1	2	3	4	4	3	1	2	4	1	3	2	3	4	3	1	2
Q.	86	87	88	89	90															
A.	1	4	4	4	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	3	2	1	3	2	2	4	3	4	2	2	3	2	1	1	1	3	1	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	2	3	1	3	2	3	3	3	4	4	2	2	2	3	1	2	4	4	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	2	2	4	4	4	1	4	3	2	2	2	3	3	2	1	2	1	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	4	3	4	2	3	3	4	4	3	4	1	4	3	4	2	4	3	4	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	3	2	4	2	3	1	3	1	2										

SOLUTIONS

PHYSICS

1) $\vec{x} + \vec{y} + \vec{z} = \vec{r}$

2) Height ascended by balloon in 2s;

$s_1 = \frac{1}{2} \times 5 \times (2)^2 = 10$ m

Velocity of balloon after 2s ;

$v_1 = 5 \times 2 = 10$ m/s

This will also be the velocity of ball when released from balloon. So further maximum height ascended by ball.

$s_2 = \frac{(10)^2}{2 \times 10} = 5\text{m}$

Total height : $s = s_1 + s_2 = 10 + 5 = 15$ m.

3)

Explain Question : Asking about the reterdation, when velocity becomes zero.
Concept : This question is based on use of calculus in general motion.
Explanation :

$v = \frac{dx}{dt} = 0 + 12 - 3t^2 = 0 \Rightarrow \quad t = 2\text{s}$

At t = 2s,

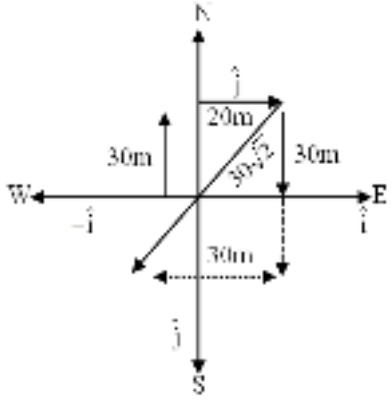
Retardation = $-\frac{dv}{dt} = -(-6t) = 12\text{m/s}^2$

Final answer : 2

4) From O to A, velocity will be positive.
At A, velocity is zero
From A to B, velocity is negative

5)

Explain Question : We have to find displacement from origin.
Concept : This question is based on displacement.
Solution :



$\vec{d} = 30\hat{j} + 20\hat{i} - 30\hat{i} - 30\hat{j}$
$= -10\hat{i}$
= 10m towards west
Final answer : 1

6)

Explain Question : We need to find the ratio of distance of magnitude of displacement in given time.
Concept : This question is based on distance and displacement.
Solution :
Given total time = 100 sec, time per lap = 40 sec
$\eta = \frac{100}{40} = 2.5 \text{ laps}$
$\frac{\text{Distance}}{\text{Dislacement}} = \frac{5\pi R}{2R} = \frac{5\pi}{2}$
Final answer : 4

7)

Explain the Question : Given $y = \frac{x^2}{2}$ and $x = \frac{t^2}{2}$ we have to calculate velocity at t = 2sec.
$[\vec{v} = v_x\hat{i} + v_y\hat{j}]$

Concept : This question is based on $v_x = \frac{dx}{dt}$
$v_y = \frac{dy}{dt}$
$\vec{v} = v_x\hat{i} + v_y\hat{j}$

Solution : $x = \frac{t^2}{2}$
$v_x = \frac{dx}{dt} = \frac{2t}{2} = t$
$y = \frac{x^2}{2} = \frac{\left[\frac{t^2}{2}\right]^2}{2}$
$y = \frac{t^4}{8}$

$$v_y = \frac{dy}{dt} = \frac{1}{8} \times 4t^3 = \frac{t^3}{2}$$

$$\vec{v} = t\hat{i} + \frac{t^3}{2}\hat{j}$$

$$\vec{v} = (t = 2) = 2\hat{i} + 4\hat{j}$$

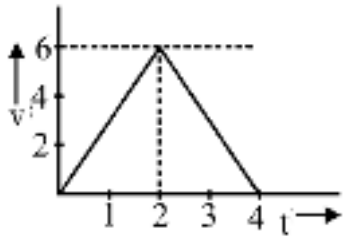
Final answer : 2

8)

$$S = \frac{\alpha\beta}{2(\alpha + \beta)}t^2$$

$$= \frac{2 \times 4}{2(2 + 4)}(3)^2 = 6m$$

9)



10)

Conceptual based

11)

Horizontal component of velocity.

12) **Explanation**

To determine the **maximum horizontal distance** the boy can throw the stone, we must consider projectile motion. The horizontal range of a projectile is maximum when the angle of projection is 45°.

Concepts Involved:

1. Projectile Motion:

• The motion of the stone can be divided into two components:

A. Vertical motion (determines the maximum height).
B. Horizontal motion (determines the range or distance traveled).

2. Formulas Used:

• The maximum height (H) is given by:

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

where:

A. u = initial velocity of the stone,

B. θ = angle of projection,
C. y = 9.8 m / s^2 (acceleration due to gravity).

• The horizontal range (R) of the projectile is given by:

$$R = \frac{u^2 \sin 2\theta}{g}$$

For maximum range, $\theta = 45^\circ$, so:

$$R = \frac{u^2}{g}$$

Step 1: Find the Initial Velocity *u*:

From the maximum height formula:

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

Here, H = 10m, $\theta = 90^\circ$ (since the stone is thrown vertically upward):

$$10 = \frac{u^2 \cdot (1)^2}{2 \cdot 9.8}$$

$u^2 = 10 \cdot 19.6 = 196$

$u = \sqrt{196} = 14$ m/s

Step 2: Calculate the. Maximum Horizontal Range R:

For the maximum horizontal range (R):

$$R = \frac{u^2}{g}$$

Substitute u = 14 m/s and g = 9.8 m/s^2:

$$R = \frac{14^2}{9.8} = \frac{196}{9.8} = 20 \text{ m}$$

Final Answer:

The maximum horizontal distance the boy can throw the stone is:
20 m

13) $R = \frac{u^2 \sin 2\theta}{g}$

14)

R tan θ = 4h

15)

Explain Question : For a projectile equation of trajectory is given & Range is to be calculated.

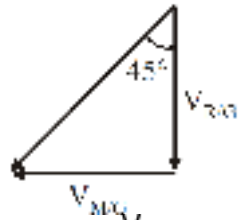
Concept : This question is based on projectile motion.

Solution : $y = 13x\left[1 - \frac{x}{39}\right]$

Comparing above equation with $y = x\tan\theta\left[1 - \frac{x}{R}\right]$
R = 39

Final Answer : option (3)

16) **Explain the Question :**
A man running with speed of 4 km/h rain apper to be falling at 45° if rain is vertical w.r.t. ground than velocity of rain.
Concept : This question is based on rain-man problem.
Solution :



$$\tan\theta^\circ = \frac{V_{M/G}}{V_{R/G}} \qquad \therefore \theta = 45°$$

$$\tan 45^\circ = \frac{5}{V_{R/G}}$$

$$1 = \frac{5}{V_{RG}}$$

V_{RG} = 5 m/s
Final Answer : 1

17) **Explain the Question :**
We have to find velocity of man such that rain drop hit him with speed of 5 m/s
Concept : This question is based on rain-man problem.
Solution :
Let velocity of man with respect to ground = $v\hat{i}$ Velocity of rain w.r.t. man $(\vec{v}_r) = (2\hat{i} - 4\hat{j})$

$$\vec{v}_{rm} = \vec{v}_r - \vec{v}_m$$
$$|\vec{v}_{rm}| = (2\hat{i} - 4\hat{j}) - v\hat{i}$$
$$|\vec{v}_{rm}| = [2\hat{i} - v\hat{j} - 4\hat{j}]$$
$$5 = \sqrt{(2-v)^2 + (4)^2}$$

$2s = (2 - v)^2 + 16$
$(2v)^2 = 9$
$2–v = ±3$
$2 - v = ± 3$
$v = 5\hat{i}$ and $-1\hat{i}$
Final answer : 4

18) **Upstream Speed**: V_b - Vw
Downstream Speed : V_b + Vw

19)

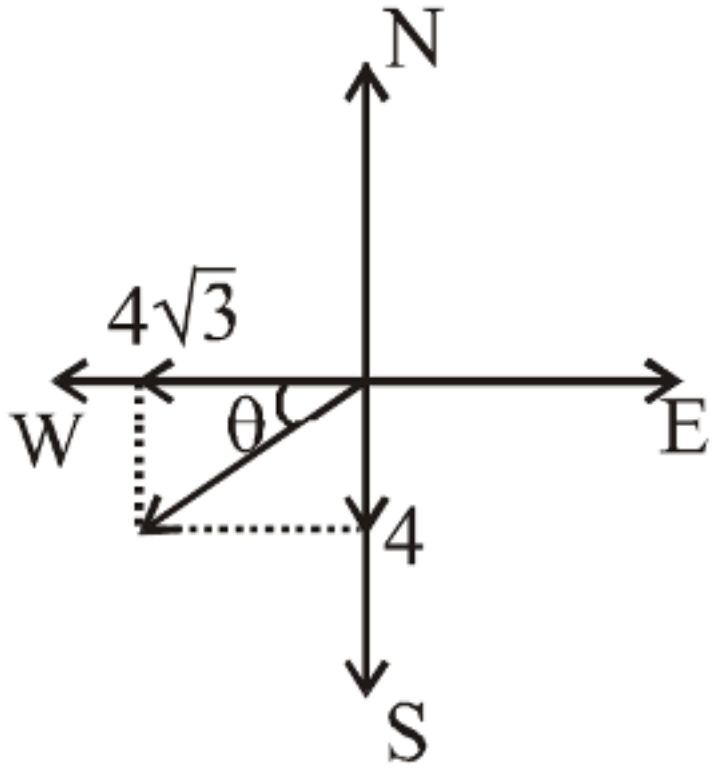
Direction based

20)

$V_{b/r} = V_b - V_r$

21) $\vec{v}_{wo} = \vec{v}_w - \vec{v}_o = \left(-4\hat{j}\right) - \left(4\sqrt{3}\hat{i}\right)$

$= -4\sqrt{3}\hat{i} - 4\hat{j} \Rightarrow |\vec{v}_{wo}| = 8$ km/hr



$\tan\theta = \frac{4}{4\sqrt{3}} = \frac{1}{\sqrt{3}}$

$\theta = 30°$ $\vec{v}_{wo} = 10$ km/hr 30° S of W

22) $\pi - \cos^{-1}\left(\frac{1}{n}\right)$

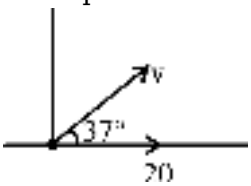
23)

v cos 37 = 200/t

v sin 37 = 30

24)

x component of velocity of both the ships are same.



v cos 37 = 20

$v \times \frac{4}{5} = 20$

v = 25

25)

an acceleration

26)

straight line

27) $\vec{V}_{avg} = \frac{vt\hat{i} + vt\hat{j}}{2t}$

$|\vec{v}_{avg}| = \frac{v}{\sqrt{2}}$

28)

Consider the motion from tree

$v_s(t_2) = v_R(t_1 + t_2)$

29) $v = 90 \times \frac{5}{18} = 25$ m/s

$t = \frac{d}{v} = \frac{75 + 425}{25} = \frac{500}{25}$

$\boxed{t = 20 \text{ sec}}$

30) $\frac{dV}{dt} = -0.2V^2$

$\int_{10}^{V} V^{-2} dV = -0.2 \int_{0}^{2} dt$

$\left[-\frac{1}{V}\right]_{10}^{V} = -0.2[t]_0^2$

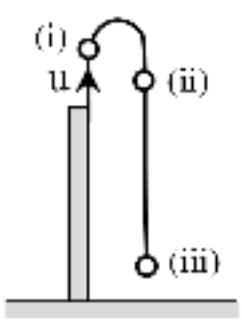
V = 2m/sec

31)

$\omega t + \theta = \pi/2$

32) $a = \frac{d^2x}{dt^2}$

33)



34)

Question Asking About: Concept and problems Approach

We are tasked with finding the total time taken by a stone to reach the ground after being dropped from a height of 125 meters. The stone is stopped and then released again after falling to a height of 45 meters. To solve this, we will use the equations of motion under gravity.

Given Data:

• Initial height from which the stone is dropped: $h_1 = 125$ m
• Height from which the stone is released after being stopped: $h_2 = 45$ m
• Acceleration due to gravity $g = 9.8$ m/s^2.
We will break the problem into two parts:
1. Time taken to fall from 125 m to 45 m.
2. Time taken to fall from 45 m to the ground.

Solution/Explanation/Calculation:

Step 1: Time taken to fall from 125 m to 45 m
The stone is dropped from rest, so the initial velocity u = 0. The height fallen is:
$h_1 - h_2 = 125 - 45 = 80$ m
We can use the second equation of motion to calculate the time taken to fall 80 m.

$$s = ut\frac{1}{2}gt^2$$

Where:
• s = 80 m.
• u = 0 (initial velocity).
• g = 9.8 m/s^2
• t is the time taken
Substituting the values:

$$80 = 0; t + \frac{1}{2} . 9.8 . t^2,$$

$$80 = 4.9\ t^2 ,$$

$$t^2 = \frac{80}{4.9} \approx 16.33$$

$$t \approx \sqrt{16.33} \approx 4.04$$ seconds.

Step 2: Time taken to fall from 45 m to the ground
Now, the stone is released from a height of 45 meters. Again, the initial velocity u = 0. We can use the second equation of motion to calculate the time taken to fall 45m:

$$s = ut\frac{1}{2}gt^2$$

Where :
• s = 45 m.
• u = 0
• g = 9.8 m/s^2
• t is the time taken
Substituting the values:

$$45=0.t + \frac{1}{2} . 9.8 . t^2.$$

$$45 = 4.9\ t^2$$

$t^2 = \frac{45}{4.9} \approx 9.18$
$t \approx \sqrt{9.18} \approx 3.03$ seconds.

Step 3: Total Time Taken

The total time taken for the stone to reach the ground is the sum of the time taken in both parts:

$t_{total} = t_1 + t_2 = 4.04 + 3.03 = 7.07$ seconds.

Conclusion:

The total time taken by the stone to reach the ↓und is approximately 7 seconds
Hence, option (3) is correct.

35) $t = \sqrt{\frac{2h}{g}}$

36)

1st and 10th second

37)

does not depend on mass

38)

Fact

39)

Average velocity = $\frac{x_f - x_i}{t} = \frac{-10-0}{15} = \frac{-2}{3}$

40)

Conceptual

41) In this graph

Slope = $\frac{dt}{dx} = \frac{1}{v}$

So, $\frac{1}{v_A} = \tan 53^\circ$

$\frac{1}{v_B} = \tan 30^\circ$

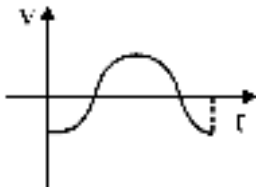
Taking ratio

we get $\frac{v_A}{v_B} = \frac{\sqrt{3}}{4}$

42)

slope of x -t curve gives instantaneous velocity

43) $v \times t$

44) 

45)

Rate of change of slope

CHEMISTRY

46) **Explain Question:** Identify which statement about electron removal/addition and molecular orbitals (MOs) is wrong.
Concept: This question is based on molecular orbital theory
Solution:
1. $N_2 \rightarrow N_2^+$: N_2 HOMO is σ2p (bonding) ⇒ removal from bonding. (True)
2. $O_2 \rightarrow O_2^+$: O_2 HOMO is π*2p (antibonding) ⇒ removal from antibonding. (True)
3. $O_2 \rightarrow O_2^-$: added electron goes into π*2p (antibonding), not a bonding orbital. (False)
4. $B_2 \rightarrow B_2^-$: B_2 HOMO is π2p (bonding) ⇒ added electron goes into a bonding π orbital. (True)
Final Answer: 3

47)

	number of unpaired electron	**magnetic moment**
O_2^-	1	1.73 B.M
O_2^-	1	1.73 B.M
O_2	2	2.83 BM

48) $POCl_3$
Side atoms different :-

49) ICl_2^-

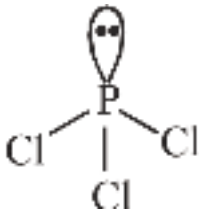


50)

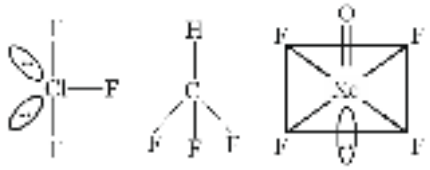
Explain Question : Identify molecule with all identical bond lengths among given options.

Concept : This question is based on bond lengths

Solution : PCl_3 has all equal bond lengths due to its trigonal pyramidal shape, making all three P-Cl bonds equivalent.

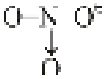


Final Answer : option (3)

51) 

52)

XeO_3F_2 ⇒ 3 pπ–dπ bonds

53) 

54) NCERT Reference: XII, Part-II, Page No. 294

55) Lone pair on equitorial position.

56)

Earlier angles were 120° and 90°.
Now 109.28º.

57)

Explanation

A. A bond angle of 90° is typically found in molecules with octahedral or square planar geometry where ligands are positioned at the corners of these shapes around a central atom.

Concept

A. 1. IF7: Iodine heptafluoride has a pentagonal bipyramidal geometry. Its bond angles are 90°, 72°, and angles close to 180°.

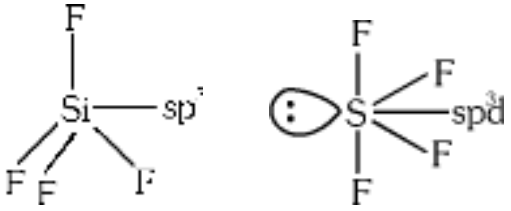
B. 2. $SiCl_4$: Silicon tetrachloride has a tetrahedral geometry. All bond angles are approximately 109.5°.

C. 3. XeF_4: Xenon tetrafluoride has a square planar geometry. It contains bond angles of 90° and 180°.

D. 4. PCl_5: Phosphorus pentachloride has a trigonal bipyramidal geometry. Its bond angles are 90°, 120°, and 180°.

Therefore, $SiCl_4$ does not contain a bond angle of 90°.Answer option 2, ($SiCl_4$).

58) **Explain the Question:** Identify the pair of species that are not isostructural.
Concept: This question is based on VSEPR theory and molecular geometry.
Solution:
SiF_4 → tetrahedral (AX_4)

SF_4 → see-saw (AX_4E) → **different geometry** 

Others [IO_3^- & XeO_3: pyramidal (sp^3); BH_4^- & amp; NH_4^+ : tetrahedral(sp^3); PF_6^- & SF_6 : octahedral(sp^3d^2)] are isostructural.
Final Answer: 1

59) $B.A \propto \frac{sAsize}{sAEN}$

60)

Explanation

A. Bond angle is determined by VSEPR theory, which minimizes electron pair repulsion around the central atom. Factors include lone pairs (increase repulsion, decrease angle), and the number of bonding pairs and their arrangement (hybridization).

Concept

A. CO_2: Carbon has 2 bond pairs and 0 lone pairs (linear geometry, sp hybridization). Bond angle = 180°.

B. NH_4^+: Nitrogen has 4 bond pairs and 0 lone pairs (tetrahedral geometry, sp^3 hybridization). Bond angle = 109.5°.

C. H_2O: Oxygen has 2 bond pairs and 2 lone pairs (bent geometry, sp^3 hybridization). Lone pair-lone pair and lone pair-bond pair repulsions reduce the bond angle to approximately 104.5°.

D. H_3O^+: Oxygen has 3 bond pairs and 1 lone pair (trigonal pyramidal geometry, sp^3 hybridization). Lone pair-bond pair repulsions reduce the bond angle to approximately 107°.

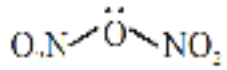
Answer option 4, CO_2 (has the maximum bond angle of 180°).

61) CH_3^+, CO_3^{-2}, BCl_3, PCl_5

62)

Question Explanation: Select the set in which every given species has bond order 2.5.
Concept: This question is based on Molecular orbital theory
Solution: O_2^+ & NO (15 electrons)
BO = 2.5 .
CN & NO^{+2} (13 e^-) also gives
BO = 2.5.
Final Answer: Option (1)

63)



at O-atom bent (sp^3)
hence polar

64) Both acetone & acetonitrile are polar substance.

65)

Both have sp^2 hybridisation.

66) B.O.

CO = 3
CO_2 = 2
CO_3^{-2} = 1.33
CH_3COO^- = 1.5

67) NCERT XIth Pg # 127, fig. 4.21

68)

NCERT XI Pg. # 113 (Part-I)

69)

Explain Question : Identify compound containing unpaired electron.
Concept : This question is based on Molecular Orbital Theory
Solution : KO_2 is potassium superoxide ($K^+O_2^-$). BaO_2 and Na_2O_2 are peroxides (O_2^{2-}). The superoxide ion (O_2^-) has 13 valence electrons, with one unpaired electron in an antibonding (π*) orbital.
Final answer : 1

70) $\underbrace{HF}_{\text{H-bonding}} < \underbrace{HI < HBr < HCl}_{\text{mol. mass } (\downarrow) \text{ volatile nature } (\uparrow)}$

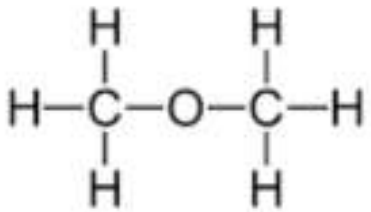
71) **Explanation** -
The question asks you to identify the molecule in which the **shown hydrogen bond is not possible**.

Concept - **(H-Bonding)**
The hydrogen bonding shown in option 3 is not practically feasible due to the para position.

Final Answer - Option (3)

72)

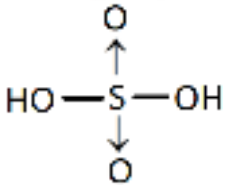
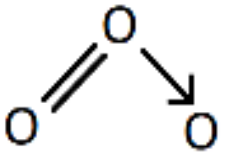
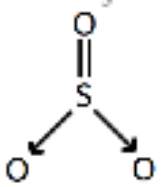
Explain Question: Identify compound lacking intermolecular hydrogen bonding.
Concept: Hydrogen bonding
Solution: Hydrogen bonding occurs when H is attached to N, O, or F. CH_3OCH_3 has H attached to C, so no intermolecular H-bonds form.



Final Answer: Option (4)

73)
Explanation → In this question we have find the compound containing co-ordinate bond.

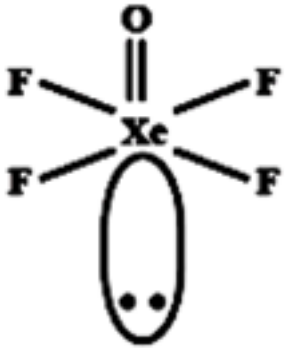
Concept → co-ordinate bond
Solution →

H_2SO_4 	O_3 	SO_3 

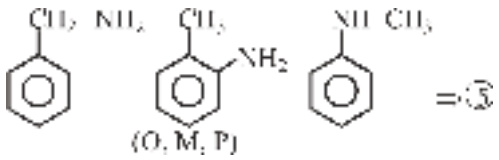
Final Answer → (4)

74) **Explain the Question :** Determines the missing values in the VSEPR table, then calculate the sum of those four variables.
Concept : This question is based on VSEPR theory.
Solution : Using VSEPR : a = 3 (Linear AX_2E_3, SN = 5). b = 4 (See-Saw AX_4E_1). c = 2 (square planar AX_4E_2,SN = 6). d = 5(T-shaped AX_3E_2). The sum a + b + c + d = 3 + 4 + 2 + 5 = 14.
Correction : Since 14 is not an option, re-evaluate. b must be 5 (see-saw AX_4E_1 has SN = 5). If b = 5, then 3 + 5 + 2 + 5 = 15.
Final Answer : 3

75) **Explanation** - The question asks which of the given molecules ($XeOF_4$, I_3^-, SF_4, XeF_5^-) exhibits sp^3d^2 hybridization.
Concept - (Hybridization- sp^3d^2)



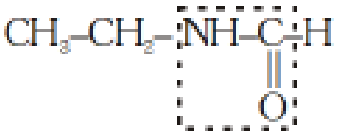
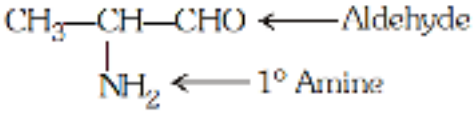
($5\sigma + 1lp = sp^3d^2$)
Final Answer - Option (1)

76) 

77)  are identical

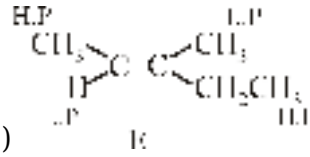
78)

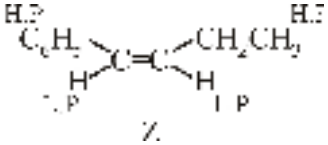
metamers are differ in alkyl groups

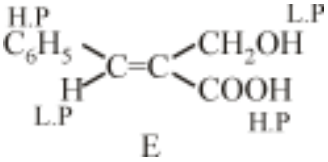
79)  (Amide functional group); 

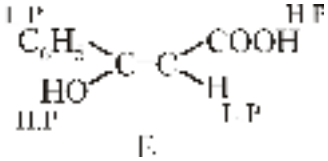
(Aldehyde and 1º amine)

80)

(1) 

(2) 

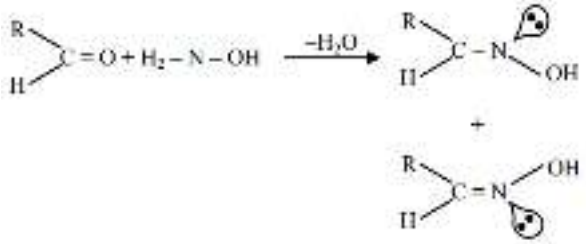
(3) 

(4) 

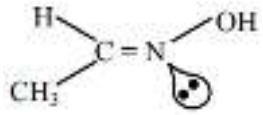
Here : H.P. → High priority group.
L.P. → Low priority group.

81)

Explaining the Question:
Identify syn aldoxime
Concept: sym-anti configuration of geometrical isomerism.
Solution: Asymmetrical Aldehydes when treated with hydroxyl amine [NH_2OH] forms aldoxime shows geometrical isomerism.



If 'H' and 'OH' are an same side : sy^{n}
If 'H' and 'OH' are on opposite side : Anti



is sy^{n} aldoxime

82)

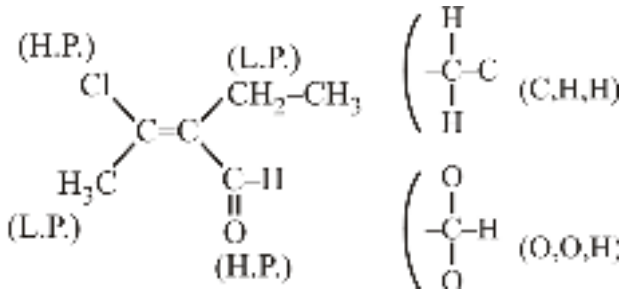
Question explanation: The question asks for the fundamental reason why geometrical isomerism occurs in molecules. We need to identify the property that leads to the existence of different spatial arrangements of atoms in a molecule.

Concept: Geometrical Isomerism (cis-trans isomerism)

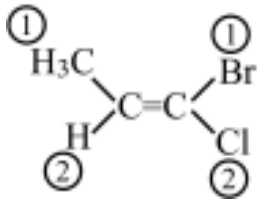
Answer Explanation:

Hindered rotation is the correct answer. Geometrical isomerism occurs when there is restricted rotation around a bond. This restriction prevents the interconversion of different spatial arrangements of atoms or groups attached to the bond.

Final Answer: Option 4 Hindered rotation.



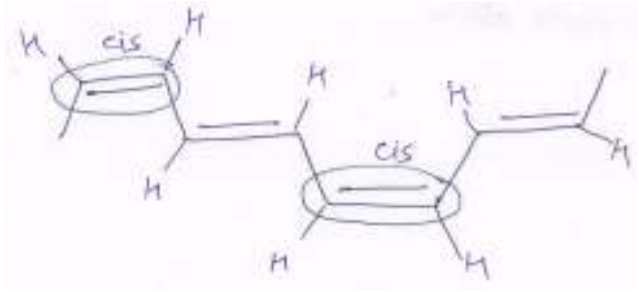
83)
E-Isomer [According to CIP rule]



84) → Z

85)

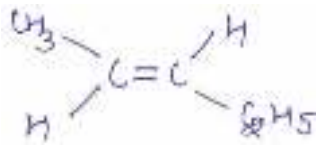
A. Cis arrangement about double bond has same groups on same side. It is shown below:



Ans 2

86)

When terminal carbon across Restricted rotation has two different valencies then it show geometrical isomer.(G.I)

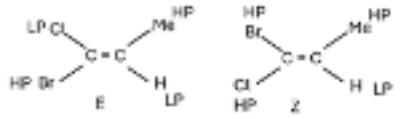

 show G.I

87)

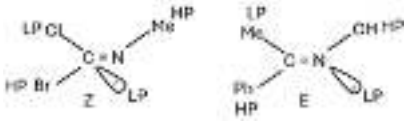
Question Asking About:

Pair of geometrical isomers

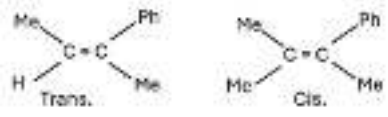
Solution/Explanation/Calculation:

(1) 

 there are pair of geometrical isomers.

(2) 

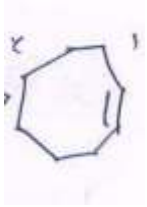
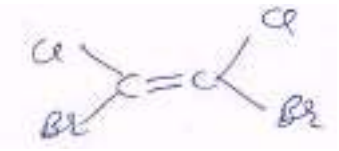
 there are pair of geometrical isomers.

(3) 

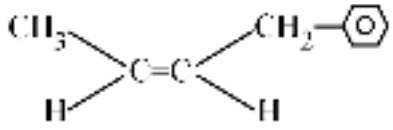
 there are pair of geometrical isomers.

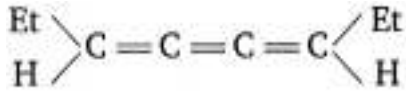
88)

* G-I- is possible in cycloalkene across endocyclic π bond if ring size is of 8 or more than 8 carbons.

 is not a planer compound. **Note:** Correction in 2nd option 

Option: 4 

89) 

90)  show geometrical isomers.

BIOLOGY

91) **Question Explanation :** The question asks you to count how many of the listed factors are external factors that affect photosynthesis in plants.
Concept:
This question is based on FACTORS AFFECTING PHOTOSYNTHESIS.

Solution:
The correct answer is 3.4

Explanation :
External factors = 4
Light, CO_2 concentration, Temperature, Water

Final answer: 3. 4

92)

The correct answer is 3: All a, b, c
Photosynthesis is:

A. Endergonic because it requires an input of energy (from sunlight) to convert carbon dioxide and water into glucose and oxygen.
B. Anabolic because it builds larger molecules (glucose) from smaller ones (CO_2 and H_2O).
C. Physico-chemical because it involves both physical and chemical processes (light energy conversion, electron transport, etc.).

93)

Question Explanation: Components of a photosystem.

Concept: Light Harvesting Complex.

Solution:

In the diagram of a photosystem, **A** represents **antenna molecules**, which absorb **B**, the incoming **photons**.
From where they are transferred to the **primary acceptor (C),**These photons energize electrons that move to the **reaction centre (D).**

Final Answer: A- Antenna molecules, B-Primary acceptor , C- Photons, D-Reaction centre

94)

Correct answer: 1. Both A & B are correct

Explanation:

A. Statement A is correct: The biosynthetic reactions (also called dark reactions or Calvin cycle) do not require light directly. They are light-independent, although they depend on products of the light reaction (ATP and NADPH).
B. Statement B is also correct: Even after light is switched off, the biosynthetic reactions continue for some time using the stored ATP and NADPH, but eventually stop when those supplies are exhausted.

So, both statements are correct.

95) NCERT (XI) Pg # 211,212,214

96) NCERT XI pg.# 212, 213

97)

NCERT-XI, Pg. # 139, 140

98) NCERT Pg. No. 139 & 145

99)

Explain Question: Identify the correct match in the table.
Concept: Factors affecting Photosynthesis.
Solution: In dense forest Light Intensity is the limiting factor for photosynthesis.
So, the correct answer is: 3.
Final answer: 3

100) NCERT (XIth) Pg. # 214

101)

NCERT (XIth) Pg. # 216, 217

102)

Explain Question: How many of the following represent the correct match of discoverer and discovery?
Concept: This question is based on Early Experiments (Photosynthesis)
Solution:
(B) Engelmann – Action spectrum of photosynthesis
(C) Julius von Sachs – Relation among photosynthesis, chloroplast, and starch.
So, the correct answer is: 2. Two.
Final answer:2. Two

103) NCERT (XIth) Pg. # 208, 209

104)

Explaining the Question: This question asks whether both statements about light reactions and naming of photosystems are correct.

Concept : Light reactions

Solution –

A. Statement-I is correct: Light reaction includes light absorption, water splitting, O_2 release, ATP & NADPH formation.

B. Statement-II is incorrect: PS-II acts first, but PS-I was discovered earlier. Numbering is based on discovery, not function.

Final Answer: Option 2 – Statement-I correct, Statement-II incorrect.

105)

A. **Statement I:** "Light is a rare limiting factor for photosynthesis in nature" is incorrect. Light can be a limiting factor for photosynthesis in certain environments, especially in dense forests or during cloudy days, but it is not always rare as a limiting factor. Factors like carbon dioxide concentration and temperature are often more limiting in many natural conditions.
B. **Statement II:** "C_4 plants show saturation at 360 ppm of atmospheric CO_2 concentration" is correct. C_4 plants are more efficient at photosynthesis under higher carbon dioxide concentrations. They show saturation (the point where photosynthetic rate no longer increases with rising CO_2) at around 360 ppm, which is close to current atmospheric CO_2 levels.

So, Statement I is incorrect and Statement II is correct.

106)

NCERT XI Pg # 143 - 146

107)

NCERT XI Pg # 216-220

108)

NCERT-XI, Pg # 215

109)

NCERT-XI, Pg. # 218

110) Allen module.

111) NCERT XI Pg. # 220

112) NCERT-XI, Pg. 210

113) NCERT-XI, Pg. # 223 (13.10.2)

114)

Allen module Page 89.

115) NCERT-XI, Page No. 219

116) NCERT-XI, Page No. 217

117)

NCERT XI, Pg. # 207

118) NCERT-XI, Pg. 209

119)

The correct answer is:
3. Release of CO: and utilization of ATP

Explanation: Photorespiration is a process that occurs when **RuBisCO** (the enzyme responsible for fixis CO_2 during the Calvin cycle) mistakenly uses **O_2**, instead of **CO_2**. This process happens more frequently under conditions of **high oxygen** and **low carbon dioxide** concentrations. During photorespiration:
1. CO_2 is released as a byproduct, rather than being fixed into sugar molecules.
2. ATP is consumed because energy is used to undo the wasteful reactions of photorespiration, and there's no direct production of sugars or NADPH.
Therefore, the correct answer is 3.

120)

NCERT-XI, Pg. # 147

121) The correct answer is **4. CO2↓, O2↑, Light intensity↑ , temperature↑.**
EXPLANATION

A. RuBisCO is an enzyme that can catalyze two reactions:
B. Carboxylase activity: This is the preferred reaction, where RuBisCO fixes CO_2 into RuBP, initiating the Calvin cycle.
C. Oxygenase activity: This is a wasteful reaction where RuBisCO fixes O_2 instead of CO_2, leading to photorespiration.
D. **Factors favoring oxygenase activity:**
E. **High 02 concentration:** When O_2 levels are high relative to CO_2, RuBisCO more likely to bind to O_2, leading to photorespiration.
F. **Low CO2 concentration:** If CO_2 levels are low, RuBisCO has a higher change of binding to O_2 instead of CO_2.
G. **High light intensity:** Increased light intensity can lead to higher O_2 production through photosynthesis, which can further favor oxygenase activity.
H. **High temperature:** Higher temperatures can also favor oxygenase activity, as it can increase the rate of photorespiration.

Therefore, the combination of low CO_2, high O_2, high light intensity, and high temperature creates conditions that favor the oxygenase activity of RuBisCO.
Therefore, the correct answer is 4.

122)

NCERT-XI, Pg. # 145,146

123)

in the Hatch & Slack (C4) pathway.
The correct answer is:
2. Bundle sheath cell and mesophyll cell
Explanation:
In the **C4 pathway** (also called the **Hatch & Slack pathway**), the process involves two main cell types: **mesophyll cells** and **bundle sheath cells**. Here's the breakdown:
1. In the mesophyll cell:

A. **CO_2** is initially fixed to **phosphoenolpyruvate (PEP)** to form a **C4 acid** (often **oxaloacetate,**

which is then converted to malate or aspartate).

2. In the bundle sheath cell:
• The **C4 acid** (like **malate**) is transported from the mesophyll cells to the bundle sheath cells.
• **Decarboxylation** of the C4 acid occurs here, releasing CO_2, which then enter the **Calvin cycle** for sugar production.
So, the **decarboxylation of the C4 acid** occurs in the **bundle sheath cell,** and the **regeneration of PEP** occurs in the **mesophyll cell**.
Therefore, the correct answer is **2. Bundle sheath cell and mesophyll cell.**

124) **Explain Question:** The question asks us to match four processes related to photosynthesis with their respective locations or outcomes
Concept: This question is based on light-dependent and light-independent reactions
Explanation:
1. Non-cyclic ETS - (b) Grana thylakoid
2. Cyclic ETS -(a) Grana and Stroma thylakoid
3. Chemiosmotic theory - (d) Formation of ATP
4. Dark reaction - (c) Stroma
Final answer: (2)

125)

NCERT, Pg. # 149

126)

Ncert Page 214 (Ncert 2022).

127) NCERT-XI, Page No 211

128)

NCERT-XI, Pg. # 208

129) NCERT (XI) Eng. Pg. # 220
NCERT (XI) हिन्दी पेज # 222

130)

NCERT Pg. # 144

131) NCERT, Pg. # 137, 138

132)

The correct answer is **3. It forms 'Z' shape when the carriers of ETS are arranged in sequence of redox potential scale.**

Explanation: The **Z scheme** is named for the "Z" shaped path formed when plotting the sequence of electron flow in photosynthesis based on the redox potentials of the electron transport chain components. The electron flow starts in Photosystem II, moves uphill, and the downhill to Photosystem I and finally reduces NADP+ to NADPH.

133) The correct answer is **24 ATP. Explanation:** In **C3 plants**, the process of synthesizing a hexose sugar (6-carbon molecule) through photosynthesis involves the Calvin cycle. It requires **18 ATP** and **12 NADPH** molecules to produce one molecule of glucose ($C_6H_{12}0_6$).
In **C4 plants,** the process is more energy-intensive. C4 plants initially fix CO_2 into a 4-carbo compound (e.g., oxaloacetate), which is then converted to malate or aspartate. This occurs the mesophyll cells. The malate is transported to the bundle sheath cells, where it releases CO2, which is then used in the Calvin cycle. This additional step requires **additional ATP** to transport the 4-carbon compound and regenerate phosphoenolpyruvate (PEP).

A. **C4 plants** require an extra **2 ATP** per molecule of CO2 fixed due to the additional steps in the C4 pathway.

To synthesize **2 molecules of hexose sugar, C4 plants** would require an additional **24 ATP** compared to **C3 plants** (12 ATP for each of the 2 hexose sugars).
Hence, the correct answer is **24 ATP**.

134)

Explain Question – The question is asking about assertion and reason.
Concept – Based on dark reaction of photosynthesis
Solution –
The assertion is true as dark reaction is called biosynthetic phase.
The reason is also true. Dark reactions depend on ATP and NADPH, not directly on light.
However, the reason does not directly explain why it's called the biosynthetic phase. The term "biosynthetic" refers to the formation of glucose, not the light independence.
Final answer – Option 2

135)

NCERT Pg. # 215

136)

Question Explanation : To identify the reason for neuron’s inability to divide

Concept: Nerve Cell - Neuron

Solution:
Centrioles are small, cylindrical organelles that play a crucial role in cell division. They are involved in the formation of the spindle fibers that separate chromosomes during mitosis (cell division). Nerve cells (neurons) in mature mammals generally do not divide because they lack centrioles.

Final answer: Option (4)

137) **Explain Question :** Schwann cells are found on
Concept : This question is based on Myelinated and non myelinated nerve
Explanation : They are mainly associated with myelinated nerve fibres, though they can also be present in unmyelinated fibres without forming a myelin sheath.
Final answer : 3 – Both (1) and (2)

138) **Explain Question :** Synaptic vesicle occurs in
Concept : This question is based on transmission of impulse
Explanation : Synaptic vesicles store neurotransmitters and are located in the axon terminal of the presynaptic neuron.
Final answer : 1 – Presynaptic neuron

139)

Explain Question : The forebrain consists of
Concept : This question is based on forebrain
Explanation : The forebrain is the largest part of the brain, including the cerebrum and diencephalon (thalamus and hypothalamus).
Final answer : 4 – Thalamus, hypothalamus and cerebrum

140)

Question Explanation : The question asks what type of function association areas of the cerebrum perform.

Concept : Functions of association areas in the cerebral cortex

Solution : Association areas do not perform purely sensory or purely motor functions. They help in complex processes like learning, reasoning, memory, and integration of information. Hence, they are neither clearly sensory nor motor.

Final Answer : Option 3

141)

142)

143)

144)

145)

NCERT-XI, Pg. # 232

146)

Question Explanation: The question checks whether the statement about the folded structure of the human brain (gyri and sulci) is justified by the given reason.

Concept: Gyri and sulci in cerebrum and cerebellum

Solution: The cerebrum is highly folded into gyri and sulci to increase surface area of grey matter. The cerebellum also has folds, but this does not explain why the cerebrum is highly folded; therefore the reason does not justify the assertion.

Final Answer: Both Assertion & Reason are true but Reason is not a correct explanation of the assertion.

147)

NCERT-XI, Pg. # 233

148)

Question Explanation: The question asks which layer forms the outermost covering of the brain.

Concept: Meninges (protective coverings of the brain)

Solution: The brain is covered by three meninges: duramater (outermost), arachnoid (middle), and piamater (innermost). Therefore, the outermost covering is duramater.

Final Answer: Duramater

149)

SOLUTION/EXPLAINATION:

Depolarization occurs during the initiation of an action potential in a neuron. Here's what happens:

A. **Resting state:** The inside of the neuron is negatively charged relative to the outside, with high concentrations of Na^+ outside and K^+ inside the cell.
B. **Depolarization phase:**

- When a stimulus reaches the neuron, voltage-gated sodium (Na^+) channels open.
- Sodium ions (Na^+) rapidly move **into the cell** down their concentration and electrochemical gradients.
- This causes the inside of the cell to become less negative (and eventually positive) compared to the outside.

Incorrect options:

- **2. Na^+ ions rapidly move outside the cell:** Sodium ions move inside, not outside, during depolarization.

- **3. K^+ ions rapidly move outside the cell:** Potassium ions move outside during **repolarization**, not depolarization.

- **4. K^+ ions rapidly move inside the cell:** Potassium ions are already inside the cell; they do not move in during depolarization.

Thus, the influx of Na^+ ions is the key event in **depolarization**.

150)

Explanation:

In a non-myelinated axon at resting potential, the distribution of Na^+ (sodium) and K^+ (potassium) ions follows these key principles:

A. The inside of the axon is more negative compared to the outside.
B. Na^+ ions are more concentrated outside the neuron.
C. K^+ ions are more concentrated inside the neuron.

The correct answer is: **Option (1)**

151)

Question Explanation : The question asks for the position and location of the hypothalamus.

Concept : Hypothalamus : location in the brain

Solution :

A. Hypothalamus is the basal (ventral) part of diencephalon.
B. It lies below the thalamus and forms the floor of the third ventricle.

Final Answer : 2

152)

NCERT Pg. No. # 234

153) **Question Explanation :** Activity of Na+-K+ pump during action potential

Concept: sub topic : Action Potential Phases

Solution:

• B (-55mV), C (0mV), and D (+10mV) represent states where the membrane potential is depolarized. During depolarization, other channels (like voltage-gated sodium channels) are opening and driving the membrane potential away from the resting potential. The Na+/K+ pump is still functioning at these potentials, but it's having to work against these other forces.

Final Answer : Option (3)

154) **Explain Question :**

Nissl's granules are found in ... and help in ...

Concept :
This question is based on Structure of neuron.

Explanation :
Nissl's granules occur in cyton and dendrites (not in axon) and are involved in protein synthesis.

Final answer : 4

155) **Explanation:** Since unipolar neurons are seen in embryos, bipolar neurons in the retina, and multipolar neurons in the cerebral cortex, Option 2 is correct.

Correct Answer: Option 2 (a-2, b-3, c-1)

156)

K^+ VGC – Close → True during depolarization (they open during repolarization).

Na^+-K^+ pump – off →The pump is **not active in depolarization**, so this is **also correct**.

Final Answer : option (3) **K^+ VGC – Close, Na^+-K^+ pump – off**

157) NCERT Pg. # 316(E)

158) NCERT(XIth) Pg # 319

159) **Question Explanation :** The question asks which labelled part acts as the coordinating centre for sensory and motor signalling.

Concept : Thalamus as the major relay and coordinating centre

Solution : In the diagram, label i represents the thalamus, which is the main coordinating centre for sensory and motor signalling.

Final Answer : Option 4

160) NCERT XI Pg # 319

161)

Question Explanation: The question is asking which structures together form the brain stem.

Concept: Components of the brain stem

Solution: The brain stem is composed of mid brain, pons and medulla oblongata, which connect the brain with the spinal cord and control vital functions.

Final Answer: Mid brain, pons and medulla oblongata.

162) NCERT-XI Pg. # 317

163)

NCERT XI Pg. # 316

164) NCERT (XI) Pg. # 317

165)

Explain Question : Find out the correct match.
Concept : This question is based on human brain – functions of different parts
Explanation :
(a) Limbic system with hypothalamus → regulates sexual behaviour ,correct
(b) Thalamus → relays and coordinates sensory & motor signals ,correct
(c) Association areas → involved in intersensory association, memory & communication ,correct
(d) Pons → fibre tracts interconnecting different brain regions ,correct
All statements are correct.
Final answer : 4. All correct

166) NCERT Pg. # 318(E)

167) NCERT Pg. # 321(E)

168)

Question Explanation: The question asks which part of the brain the cerebral aqueduct passes through.

Concept: midbrain.

Solution: The cerebral aqueduct connects the third and fourth ventricles and runs through the midbrain.

Final Answer: Midbrain

169)

Question Explanation: The question asks which part of the peripheral nervous system carries impulses from the CNS to the viscera and from the viscera back to the CNS.

Concept: peripheral nervous system

Solution: The system that includes nerves, fibres, ganglia and plexuses carrying impulses to

and from viscera is the visceral nervous system. It includes both motor (autonomic) and sensory components supplying internal organs.

Final Answer: Visceral nervous system

170) Pupillary dilation, stimulation of secretion of sweat, ejaculation are function of sympathetic nervous system.

171) Nissl's granules are absent in axoplasm.

172) **Question Explanation :** Labeling parts of a neuron

Concept: Neural Tissue : Neuron Structure

Solution:
Let's break down the components of the neuron:
• **Dendrites (a):** These are short, branching fibers that receive signals from other neurons.
• **Cell Body (b):** This is the main part of the neuron, containing the nucleus and other organelles.
• **Schwann Cell (c):** These cells wrap around the axon, forming the myelin sheath.
• **Axon (d):** This is a long, slender projection that transmits signals away from the cell body.
• **Myelin Sheath (e)**: This is a fatty layer that insulates the axon and speeds up signal transmission.
• **Node of Ranvier (f):** These are gaps between segments of the myelin sheath.

Final answer : **3**

173) Module

174) Module

175) NCERT Page no 105

176) NCERT-XI, Pg. # 320

177)

Module 2 Pg. # 89, 91

178) **Question Explanation –** The question asks whether the statements about Schwann cells and their function are true and whether the reason explains the assertion.

Concept – myelinogenesis

Solution –

Assertion: True. Schwann cells are present in both myelinated and unmyelinated axons of the PNS.
Reason: False. Schwann cells do NOT form myelin in the CNS.
Myelin in CNS is formed by oligodendrocytes, not Schwann cells.

So the assertion is correct, but the reason is incorrect.

Final Answer – Option 3

179) NCERT Pg#320

180) NCERT, Pg # 231